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Research Infrastructure in the Post-War Recovery: From Rescue to Growth

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LIST OF ABBREVIATIONS

VRU – Verkhovna Rada of Ukraine
RI – research infrastructure
R&D – research and development
HEI – higher education institution
CMU – Cabinet of Ministers of Ukraine
MESU – Ministry of Education and Science of Ukraine
NAASU – National Academy of Agrarian Sciences of Ukraine
NAMSU – National Academy of Medical Sciences of Ukraine
NASU – National Academy of Sciences of Ukraine
RDW – research and development works
SI – scientific institution
NRFU – National Research Foundation of Ukraine
UANIPIO – Ukrainian National Office for Intellectual Property and Innovations
CSUSE – Centre for shared use of scientific equipment
CEB – central executive bodies

INTRODUCTION

Research infrastructure (RI) plays an ultimate role for maintaining and developing a country's scientific and innovation potential. RI comprises specialised facilities, equipment, and laboratories necessary for conducting high-quality scientific research. Simultaneously, their functioning contributes to the emergence of new ideas, technologies, and innovations, which in its turn can lead to the creation of new products, services, and jobs. International cooperation in construction, maintenance, and modernisation of RI, as well as direct usage of RI by scientists and knowledge-intensive enterprises alike, can facilitate the exchange of knowledge, experience, and technologies. The EU's science and innovation policy has a special place for RI and regards it as centres of knowledge and the foundation for economic competitiveness[1].

Researchers in academia, universities, and business need access to modern RI to be able to conduct cutting-edge research. Access to RI is especially crucial for (I) young researchers as part of their academic education and career start, and (II) for companies that typically cannot afford to purchase their own measuring, testing, and research equipment due to lack of financial resources[2].

RI can be viewed as strategic assets of the state, which are important for providing research communities with resources and services to conduct research and promote innovation. Therefore, a strategic approach to the development of RI is essential for the sustainable development of science, support of skills, competences, and knowledge, as well as ensuring socio-economic development and innovation potential of scientific institutions and high-tech businesses.

The RI system at the level of a particular country plays a significant role in the context of the national innovation system, which consists of elements and interconnections principal for the production, dissemination, and use of new and economically useful knowledge[3]. Therefore, strategic planning of RI investments,

[1] Research Infrastructures as knowledge hubs and pillars of competitiveness. ESFRI. URL: *https://www.esfri.eu/esfri-white-paper/11-research-infrastructures-knowledge-hubs-and-pillars-competitiveness*.
[2] Final Report of an Independent European Audit of Ukraine's National Research and Innovation System. 2017. URL: *https://op.europa.eu/en/publication-detail/-/publication/adb1f077-dedc-11e6-ad7c-01aa75ed71a1/language-uk*
[3] Nelson, 1993; Lundvall, 1992.

implementation of effective governance and management practices, as well as openness and inclusion of national RI in international networks are important for the academic potential of a particular country. Despite limited resources, both developing and developed countries pay particular attention to maintaining and developing their own RI networks through a widespread process of thorough inventory, mapping[4], and prioritization[5].

During various kinds of crises — natural disasters, man-made emergencies, military conflicts —RI suffers the most and requires considerable resources for recovery. In Ukraine, the task of its restoration is greatly complicated by the fact that it must be carried out in the context of the ongoing armed conflict in order to continue scientific research.

Russia's military aggression against Ukraine largely targets civilian infrastructure, including research facilities; according to a March 2023 report by the Ministry of Education and Science of Ukraine, about 35% of research facilities were damaged or destroyed[6]. Additionally, even before the full-scale invasion, Ukraine had a number of problems in the field of research funding, availability of modern equipment, and coordination of efforts between different research institutions. Unless these problems are resolved, they will form a restrictive factor for the recovery of science after the crisis and the country's ability to rebuild "better than before". The authors believe that the effective functioning of the existing RI system needs to be ensured in Ukraine, making the most of the opportunities for international cooperation and integration into the European Research Area (ERA) to ensure participation of science in the post-war recovery of the country.

This research paper provides recommendations for restoring and modernising Ukraine's research infrastructure system. The main issue that this paper attempts to find solution for is the fact that the outdated, damaged, and destroyed research infrastructure and inefficient organisational system of its management greatly limit the development of science and post-war recovery of Ukraine. This document presents the specifics about the functioning and typology of RI, the main issues in this field in Ukraine, and also provides suggestions for the development and operation of the RI system.

The paper comprises four sections aimed at analysing the RI system both conceptually and practically. Section 1, which draws on the EU experience, discusses the RI system key elements, including its operational and strategic management, financing, and organisation at different levels. Section 2 provides an overview of the current status of the Ukrainian RI system. It provides an analysis of its components, including their legal form, statistics, and governance, and highlights key issues. Section 3 discusses the international experience of RI recovery in the countries that have experienced crisis situations, considering regional innovation development and smart specialisation as possible approaches to the RI development strategy. Moreover, the experiences and ideas of Ukraine are analysed based on interviews and surveys. Section 4 provides recommendations for improving Ukraine's RI system to address the identified problems drawing on Ukrainian and international experience.

[4] Caliari T., Rapini M.S. & Chiarini T. Research infrastructures in less developed countries: the Brazilian case. Scientometrics. 2020. № 122. Pp. 451–475. URL: *https://doi.org/10.1007/s11192-019-03245-2*.
[5] Rwandan Research and Implementation Writing Group: Building health research infrastructure in Rwanda. Lancet Global Health. 2014. # 2. e9–e10. URL: *https://www.thelancet.com/journals/langlo/article/PIIS2214-109X(13)70144-X/fulltext*.
[6] ERAC Report on the role of Research and Innovation in Ukraine's Recovery. 2023. URL: *https://data.consilium.europa.eu/doc/document/ST-10034-2023-INIT/en/pdf*.

METHODOLOGY

The task of analysing the RI landscape in Ukraine was carried out by a group of six people.

The work started with the definition of the objectives and the development of the research methodology. This was followed by a review of the literature, RI development guidelines and related legislation. The authors analysed statistics on the functioning of RI in Ukraine and performed case studies of the best RI operation practices, as well as cases of RI recovery after various crises.

There were several iterations in the process of formulating the research questions. To supplement information from existing publications, in-depth interviews were conducted with individuals involved in RI operation in Ukraine, as well as in the development and implementation of relevant components of state policy in the academic field. A total of 21 interviews and 2 focus groups with representatives of the HEI/SI CSUSE and representatives of displaced HEIs/SIs were conducted between 10 August and 15 September 2023. The results of these interviews and focus groups were arranged and interpreted in the context of RI issues. Some of those interviewed chose to remain anonymous.

The focus group was attended by HEI and SI representatives who were involved in the work at centres for shared use of scientific equipment:

- Experimental and applied physics laboratory (Uzhhorod National University);
- Technical and Environmental Safety of the Carpathians (Ivano-Frankivsk National Technical University of Oil and Gas);
- Experimental complex for scientific research of water management facilities (National University of Water and Environmental Engineering, Rivne);
- Innovative geoenergy (Dnipro University of Technology);
- Accelerator mass spectrometry (Institute of Applied Physics of the National Academy of Sciences of Ukraine, Sumy);
- Laboratory of materials science of solar energy, sensor and nanoelectronic systems, Centre for Biomedical Research (Sumy State University);
- Centre for Shared Use of Scientific Devices/ Equipment and Research of Biological Substances (O.V. Palladin Institute of Biochemistry of the National Academy of Sciences of Ukraine, Kyiv).

SECTION 1. ORGANISATION AND OPERATION OF THE RESEARCH INFRASTRUCTURE IN THE EU MEMBER STATES

As defined by the European Commission, research infrastructures are facilities, resources or services that are unique for research and acceleration of innovation in all fields of science. They include core scientific equipment (or sets of it), knowledge-based resources such as biological collections and archives, clean rooms, nature reserves, high-speed computer networks, e-infrastructures, research vessels and satellites, accelerators and synchrotrons, observatories, telescopes, and more[7]. It is worth noting that research infrastructures can be undistributed, distributed, networked, mobile, and virtual.

The European Union (EU) and its member states share competencies[8] in research and technology. The EU's RI development policy discourse is shaped by the European Strategy Forum on Research Infrastructures (ESFRI). ESFRI is an advisory body to the European Council that includes representatives from all EU members and associated countries (the latter without a casting vote) at the level of governmental organisations responsible for the country's scientific and technological development. ESFRI has various working groups in specific areas. Every few years, the Forum publishes the European Roadmap for Research Infrastructures identifying opportunities for coordination in financing, governance, and organization of RI by individual member states. The main RI subject areas, as defined in 2021, include[9] data, computing and digital research infrastructures, energy, environment, health and food, physical sciences, and engineering, as well as social and cultural innovation.

[7] Regulation (EU) No 1291/2013 of the European Parliament and of the Council. 2013.
[8] Article 4 of the TFEU. The EU and its Member States are able to legislate and adopt legally binding acts. Member States exercise their own competence where the EU does not exercise, or has decided not to exercise, its own competence.
[9] Landscape Analysis. ESFRI Report on Research Infrastructures. Roadmap 2021. URL: *https://roadmap2021.esfri.eu/landscape-analysis/section-1/*.

Additionally, each country provides its own clarifications at the national level in accordance with the overall strategy of economic, scientific, and technological development[10]. For instance, when national roadmaps were formed in the Czech Republic for 2010 and 2016, research infrastructures were defined as those that are unique, provide open access to the national and international community, have a significant impact on national development, and align with similar international RI. In Hungary, there is a clear distinction between "registered RI" and "RI of strategic importance". The latter is crucial for research activities, resolving tasks of national strategic importance, and conducting research of international significance, which includes providing transparent access for research groups from other countries. Bulgaria and Croatia base their definition of RI on the ESFRI definition, as supplemented with a clarification of unique RI — highly specialised infrastructures that have no analogues at the national level and are included in the European RI Roadmap.

Therefore, the ESFRI programme and action plan provide EU countries with guidelines and principles for the development of their RIs in co-operation with other countries and for the planning of public expenditure on their establishment and maintenance. Each country's specific RI development plan may differ from the above-mentioned formulations, considering priority research areas of the given country.

The objects comprising RI should primarily be viewed as strategic tools that establish the basis for research and serve as its instruments. Another important function is staff training. Infrastructure alone is not effective without the support of competences; in other words, without people equipment may remain just equipment. Thus, competences and skills have key importance for the establishment, use, and conduct of research through RI. RI and other types of competence and knowledge centres have a special place in the context of the Smart Specialisation Strategy (S3), which is an approach that identifies strategic areas and potential of the economy, taking a broad view of innovation, including technological approaches with monitoring mechanisms[11].

As separate entities, RIs are said to form a distinct organisational sphere[12] and share some common features of organisational culture[13]. For instance, publicly funded RIs (at the expense of taxpayers) are available to third-party users on a competitive basis. Users are, consequently, the key individuals and groups stimulating the RI life cycle. Additionally, RIs do not appear de novo but are often the result of a well-established scientific, technical, and engineering base.

Another RI feature is that over the past twenty years research activities involving RI have been linked to innovative outputs (products). As a result, policy discourse surrounding RI, specifically in the EU context, has focused on the socio-economic benefits that can be generated by investing in research facilities and equipment. Consequently, RI is often seen as the material basis for potential innovations arising from applied research. GRID (technologies and distributed resources used for distributed computing, modelling, and data processing), the Internet, or innovations generated in space exploration and materials science are often cited as examples. Justification for establishing a new RI or

[10] Materials of the Capacity-Building Training for Evaluation, Planning, Monitoring, Impact Management and Management of Research Infrastructures, held on 13-16 February 2018 under RESINFRA@DR project of the Interreg programme of the Danube Region countries.
[11] What is Smart Specialization? Smart Specialization Platform. URL: *https://s3platform.jrc.ec.europa.eu/what-we-do*.
[12] Hallonsten, O. (2020). Research Infrastructures in Europe: The Hype and the Field. European Review, 28(4), 617–635. URL: *https://doi.org/10.1017/S1062798720000095*.
[13] Latour, B. & Woolgar, S. (1979). Laboratory life: the social construction of scientific facts. Sage Publications. URL: *https://www.worldcat.org/title/laboratory-life-the-social-construction-of-scientific-facts/oclc/4775088?page=citation*.

continued funding and modernisation of an existing RI is typically based on the potential benefits they can bring.

The sustainability of research infrastructure has become a topic of political discussion, alongside new concepts of public management. In individual countries, budgets for RI are typically determined based on established priorities. RI mapping is a common practice in the EU, occurring at both the EU-wide and individual country levels when each country maintains a list of its own RI capabilities, enabling researchers and decision-makers to identify complementarities with other countries and support investments in those areas and RI which are considered strategic for a specific country.

It is essential to monitor the achievement of the RI objectives using key performance indicators. Additionally, RI can be evaluated through research output indicators[14] such as high-impact journal articles, the number of patents, and more[15]. However, the problem is that the complete innovative results of using RI may not align with the timeframe, such as, for instance, short-term state budgets. Meanwhile, the ability to demonstrate partial innovation results might meet the budget cycle needs.

It should be noted that monitoring and evaluation of RI differ from standard evaluation procedures for research projects with much shorter implementation periods, different objectives, and funding needs[16]. Therefore, the current status of most RIs should be updated periodically during the 10-year implementation cycle and operational phase. For this purpose, ESFRI provides the European Commission with reference information on the RI development progress included in the roadmap, and their achievement of the target technology readiness level (TRL), enabling them to apply for the next funding phase.

When assessing a RI, it is necessary to consider its life cycle phase. Each RI of national and international (regional, European, or global) significance has six life cycle stages[17]:

- concept development (screening, formation of an international consortium, coordination of access and funding policies, definition of scientific and project management formats);
- establishing the RI (including design, obtaining political and financial support, overall cost planning, development of shared access policies, management and staff recruitment);
- preparation phase (construction planning, securing political and financial support, planning data handling and management policies, and preparing estimates);
- implementation (organising the construction and location of sites, staff recruitment, intellectual property protection, planning for long-term operations and modernisation);
- operational phase (conducting research, providing services to the academic community, expanding the customer community and attracting new customers, modernising tools, techniques and methods, ensuring sustainable operation);
- termination (reconstruction, dismantling and replacement of equipment with renovation, reuse, etc.).

RI is integrated into public administration systems. Over the past two decades, especially in the context of EU and ESFRI, as well as through discourse in other international forums, such as OECD Global Science Forum[18,19] and G7[20], RIs have been classified as policy-making

[14] Kolar, J., Cugmas, M., Ferligoj, A. (2019). Towards Key Performance Indicators of Research Infrastructures. URL: *https://arxiv.org/ftp/arxiv/papers/1910/1910.00304.pdf*.
[15] Mayernik, M.S., Hart, D.L., Maull, K.E. and Weber, N.M. (2017). Assessing and tracing the outcomes and impact of research infrastructures. Journal of the Association for Information Science and Technology, 68: 1341–1359. URL: *https://doi.org/10.1002/asi.23721*.
[16] Supporting the Transformative Impact of Research Infrastructures on European Research. European Comission_B-1049 Brussels. 2020. Pp. 31–32.
[17] Public Roadmap 2018 Guide (2016). European Commission.

entities and consolidated within a specific organizational field based on their management styles, procedures, and traditions they uphold.

At the EU level, individual countries often practice joint investments in the construction and operation of RI due to their limited budgets. Typically, countries invest in networks of national RIs and become members of international RIs to support progress in their strategically important areas of science and economy. The idea that science is "policy by other means"[21]. While this statement better reflects the arms race and the Cold War era, today's global economic competition is also based on the idea that success in knowledge-intensive industries and the application of scientific advancements (such as the invention of vaccines during the pandemic[22] or the global microchip race[23]) are paramount. Simultaneously, the EU possesses resources and instruments to stimulate cooperation in research and innovation, including RI. For instance, the Regional Development Fund and loans from European investment banks are often viewed as potential additional sources of funding, particularly in the economically lagging countries and regions.

It is essential to have a strategic vision of the creation, regular upgrade, and eventual decommissioning of RI in the form of a RI life-cycle perspective, as well as its sustainable development. Besides the material and technical aspects of building and maintaining the sustainability of RI, special attention should be paid to the quality of human capital. This includes education, support, and development of competencies and skills of RI users. Ensuring that knowledge-intensive enterprises use their RI resources creates the necessary conditions for achieving innovative results with the appropriate equipment.

In conclusion, the summary of the above characteristics demonstrates that RI may be distinguished as a strategic asset for both the research community and industry of individual countries. A reliable foundation of advanced knowledge, held by skilled individuals (which means competitive human capital), and access to modern equipment to use the said base create the necessary preconditions for generating innovations beneficial to the society and supporting the country's socio-economic development.

[18] OECD Global Science Forum (2008). Report on Roadmapping of Large Research Infrastructures. URL: *https://www.oecd.org/sti/inno/41929340.pdf*.
[19] OECD (2014). International Distributed Research Infrastructures. URL: *https://www.oecd.org/sti/inno/international-distributed-research-infrastructures.pdf*.
[20] The Group of Senior Officials on Research Infrastructures (GSO). URL: *https://www8.cao.go.jp/cstp/kokusaiteki/g7_2023/annex4_gso.pdf*.
[21] Seguin, E., Lord, L.-O. (2023). Bruno Latour's Science Is Politics By Other Means: Between Politics and Ontology. Perspectives on Science, 31(1), 9–39. URL: *https://doi.org/10.1162/posc_a_00579*.
[22] Agarwal, R., Gaulé, P. (2021). What Drives Innovation? Lessons from COVID-19 R&D. URL: *https://www.imf.org/en/Publications/WP/Issues/2021/02/20/What-Drives-Innovation-Lessons-from-COVID-19-R-D-50096*.
[23] The global microchip race: Europe's bid to catch up. URL: *https://www.ft.com/content/b31e27fd-0781-4ffd-bb69-9af985abff41*.

SECTION 2. RESEARCH INFRASTRUCTURE OF UKRAINE

2.1 Legal regulation for research infrastructure

In the area of legal regulation of RI, one of the important highlights was the adoption of the Law of Ukraine *On Scientific and Technical Activities*[24] in 2015, in which lawmakers corrected the shortcomings of the previous principal law and defined research infrastructure as "a set of facilities, resources, and related services used by the scientific community to conduct research at the highest level, covering the most important objects of scientific equipment and facilities or sets of instruments, knowledge-based resources (collections, archives, depositories or data banks of scientific information), infrastructure based on communication technology (grids, computers, software, and networking), and other structures of a unique nature. Research infrastructures may be local, virtual, or distributed (an organised resource network), public or private. RIs may be part of international networks"[25].

This definition of RI is close to the one adopted in the EU. Besides, under this law, the State prioritises "integration of the national research area into the European Research Area by implementing its priorities, particularly aligning the strategy for creation of the state RI with the European RI roadmap", and ensures "the creation of a modern scientific infrastructure and information support system for scientific and technical activities, the integration of education, science, and industry."

[24] Law of Ukraine On Scientific and Technical Activity. The Official Bulletin of the Verkhovna Rada. 2016. #3. P. 25.
[25] Law of Ukraine On Scientific and Technical Activity. The Official Bulletin of the Verkhovna Rada. 2016. #3. P. 25.

This law also delineates the responsibilities of certain governance entities in relation to RI. The National Council of Ukraine for Science and Technology Development may prepare and submit proposals on the principles of establishment and strategy of development of the state research infrastructure and the system of state key laboratories. The Ministry of Education and Science of Ukraine (MESU) carries out regulatory and legal regulation in the areas of education and science, academic, scientific and technical activities, as well as innovation activities in these areas and technology transfer; it defines priority areas for the development of science, technology, and innovation activities based on long-term and medium-term forecasts for the scientific, technical, and innovation development[26]. The powers of local executive authorities in scientific and technical activities include promotion of the development of scientific and technical infrastructure in the region.

According to the same law, the main activities of state scientific institutions funded by the state budget, scientific research conducted by universities, academies, and institutes are funded from the stage budget, which is also the funding source for the development of the infrastructure of scientific and technical activities.

An important step in the development of RI was the creation of the NRFU, which aims to develop RI in Ukraine and integrate it into the global RI, as well as to facilitate scientific and technical cooperation between research institutions, higher education institutions, and representatives of the real economy sector and service sphere.

On 14 April 2021, the Government approved the Concept of the State Target Programme for the Development of Research Infrastructures in Ukraine until 2026[27] to create the necessary organisational, legal, and financial preconditions for the development of the RI system in Ukraine, improve its competitiveness, and develop the scientific and innovation sphere in terms of scientists' access to advanced research infrastructures in the EU. The draft State Target Programme for the Development of Research Infrastructures in Ukraine until 2026[28] has been developed by MESU but not adopted yet.

The implementation of the Concept was scheduled for 2021–2126. The first (preparatory) stage (2021–2022) of the programme included the development of regulations and procedures for the creation, implementation, modernisation, and operation of RIs; the establishment of a relevant committee; and the creation of instruments to financially support the participation of Ukrainian RIs in international RIs associations.

The second stage (2022–2026) envisaged the creation and operation of various organisational and legal forms of RI.

The Concept is aimed at implementing a set of activities to restore and develop the RI system and developing a mechanism for consolidating the efforts of the CEB, the National Academy of Sciences of Ukraine, national sectoral academies of sciences, and international organisations providing international technical assistance for the restoration and development of the RI system in Ukraine.

[26] Resolution of the Cabinet of Ministers of Ukraine No.630 of 16 October 2014, On Approval of the Provision on the Ministry of Education and Science of Ukraine. URL: *https://zakon.rada.gov.ua/laws/show/630-2014-%D0%BF#Text*

[27] On Approval of the Concept of the State Target Programme for the Development of Research Infrastructures of Ukraine until 2026. Order No.322-p of the Cabinet of Ministers of Ukraine of 14 April 2021. URL: *https://zakon.rada.gov.ua/laws/show/322-2021-p#n9*

[28] Draft CMU Resolution On Approval of the Concept of the State Target Programme for the Development of Research Infrastructures of Ukraine until 2026. URL: *https://mon.gov.ua/ua/news/mon-proponuye-do-gromadskogo-obgovorennya-proyekt-postanovi-kmu-pro-zatverdzhennya-derzhavnoyi-cilovoyi-programi-rozvitku-doslidnickih-infrastruktur-v-ukrayini-na-period-do-2026-roku*.

The main objectives of the Programme are:

- taking inventory and systematisation of RI;
- assessment of their compatibility with the European RI;
- providing priority funding for their further development and alignment of the Programme with the roadmap for the development of European research infrastructures approved by the European Strategic Forum for Research Infrastructures (ESFRI).

To help achieve the Programme goal, an Organising Committee for RI Development is proposed to be established.

Typically, state target programmes contain time-bound stages of their implementation in the format of individual tasks, indicating responsible executors at the institutional level, the timeframe for completion of each task, expected results with specific indicators and required resources. Such programmes are coordinated with all executive authorities involved and other stakeholders, which may result in conflicts of interest between stakeholders or a lack of resources when the state cannot guarantee their necessary amount. As the understanding of the importance of RI in Ukraine is lacking, specifically at the level of heads of relevant ministries, there have been serious difficulties with the Programme coordination. A representative of MESU noted in an interview that the full-scale aggression of the Russian Federation against Ukraine, among other things, prevented the adoption of the act.

In 2021, amendments were made to the Procedure for the Use of Funds Assigned in the State Budget for the Fulfilment of Ukraine's Obligations in International Scientific, Technical, and Educational Cooperation[29], according to which the special fund received from the relevant external EU assistance instrument is forwarded, *inter alia*, for the development of RI. They can also be used for the measures defined in the Concept of the State Target Programme for the Development of Research Infrastructures in Ukraine until 2026, including the purchase of equipment and materials for scientific research in accordance with the priorities of the EU Framework Programme for Research and Innovation (now Horizon Europe) for HEIs/SIs to stimulate their participation in it based on the results of a tender held for scientific and technical projects.

[29] On Amendments to the Procedure for the Use of Funds Assigned in the State Budget for the Fulfilment of Ukraine's Obligations in International Scientific, Technical, and Educational Cooperation.
Resolution No.896 of the Cabinet of Ministers of Ukraine of 26 August 2021. URL: *https://zakon.rada.gov.ua/laws/show/896-2021-n#text*.

2.2. Typology of research infrastructure in Ukraine

The Law of Ukraine *On Scientific and Technical Activity* defines that the RI in Ukraine includes centres for shared use of scientific equipment, national research centres, state key laboratories, and scientific facilities which are classified as national heritage (Table 1).

Table 1. RI typology in Ukraine[30]

Type of RI	Legal form of organization	Purpose of establishment	Quantity
Centres for shared use of scientific equipment (CSUSE)	Established in the form of a structural unit of a research institution, higher education institution, or independent legal entity regardless of departmental subordination and form of ownership	Providing access to unique equipment for scientific research and scientific and technical (experimental) developments by domestic and foreign scientists	102 (lost due to war – 6[31])
National scientific center	A status that can be granted to a scientific institution, university, association of scientific institutions and/or universities that have unique research and experimental equipment, scientists, and highly qualified specialists whose research results are of national importance and have been internationally recognized	Implementation of the areas of science and technology development and/or innovation activities most important and relevant for the state	14[32]
State key laboratory	It is formed in a certain scientific field. It may be a new legal entity or an association of scientific groups of institutions and/or universities, academies, institutes based on joint scientific activity agreements pursuant to the procedure established by law. The status of a state key laboratory is granted from the date of its state registration by MESU for a specified period (from five to seven years), which may be extended pursuant to the procedure established by the CMU based on the results of the key laboratory's activity report.	Support and development of global-class fundamental research, new promising interdisciplinary areas of scientific research and scientific and technical development, coordination of joint activities and efficient use of financial, logistic, and human resources	0 (1 — existed in 2011-2013 in a somewhat different format[33])
Scientific facility which is classified as national heritage	A unique facility that cannot be reproduced, and the loss or destruction of which will severely affect the development of science and society. This status is granted by CMU upon submission by MESU.	Preservation of facility and its use for scientific purposes	147[34]

[30] Law of Ukraine On Scientific and Technical Activity. Official Bulletin of the Verkhovna Rada of Ukraine. 2016. #3. P. 25.
[31] Research infrastructures. Ministry of Education and Science of Ukraine. URL: *https://mon.gov.ua/ua/nauka/nauka/derzhavni-klyuchovi-laboratoriyi. Centres for shared use of scientific equipment of NASU. NASU. https://www.nas.gov.ua/SharedResources/UA*
[32] YouControl — an online company verification service. National scientific centre. URL: *https://cutt.ly/jwYfSw8A*
[33] Order No.6 of the State Agency for Science, Innovation and Information On the Establishment of Scientific and Education Centre "State Key Laboratory of Molecular and Cell Biology of 1 April 2011. URL: *https://ips.ligazakon.net/document/FIN64116*
[34] Register of scientific facilities classified as national treasure. URL: *https://lookerstudio.google.com/u/0/reporting/ebf7d17a-e0a2-4789-bb0f-bca52abc8d38/page/p_z7na0uts3c?s=gJhT8goE-5Y*

Examples of functioning and problems of CSUSE reporting to different departments are discussed further in Section 2.5.

A national scientific centre is only a status that contains an image component and does not provide for separate funding instruments. There are 14 national scientific centres in Ukraine, most of which report to the National Academy of Agrarian Sciences — 7 centres. The Institute of Mechanisation and Electrification of Agriculture is the first national scientific centre (status since 2000).

The National Antarctic Science Centre (NASC, *http://uac.gov.ua/*), a public institution, was established in 1993 as part of MESU. It coordinates Antarctic research, ensures the operation of the Vernadsky Research Base, the Ukrainian Antarctic research station, and organises annual Antarctic expeditions. The 28th Ukrainian Antarctic Expedition is currently taking place for the year 2023–2024. NASC represents Ukraine at the annual Antarctic Treaty meetings. Ukraine is one of 29 states that hold the highest consultative status with a voting right and veto in decision-making.

NASC carries out its scientific activities within the framework of the State Target Scientific and Technical Programme of Research in the Antarctic for 2011–2023[35]. This programme provides for fundamental and applied scientific research in the Antarctic, ensuring the effective functioning of the Vernadsky Research Base, fulfilling Ukraine's international obligations under the Antarctic Treaty, and conducting a scientifically based assessment of the biological and mineral resource potential of the region. NASC ensures continuous monitoring observations and original research and fulfils international obligations of data transferring (geomagnetic, ozonometric, meteorological, and more) to international centres and organisations. NASC also creates databases for monitoring environmental parameters, compiles Photo-ID catalogues of whales, updates Weddell seal catalogues, identifies living organisms by what is called "environmental DNA", and conducts microbiological studies of Antarctic soils.

In 2024, the Vernadsky Research Base and the Noosphere icebreaker (the former Royal Navy flagship) will join the international network of polar research infrastructures consisting of 50 organisations and 79 infrastructure facilities in both polar regions—the Arctic and Antarctic (38 research stations, 12 vessels, 18 observatories, 4 repositories, 7 databases), which will allow Ukraine to attract additional funds to the Antarctic programme, and Ukrainian scientists to use polar facilities of other countries and participate in important scientific initiatives. These activities are envisaged by the POLARIN: POLAr Research Infrastructure Network project, which won a competition in the EU's Horizon Europe programme to be implemented in 2024–2025[36].

A pilot project to create a key laboratory, the State Key Laboratory of Molecular and Cell Biology Research and Training Centre, was implemented in Ukraine in 2011–2013 by the State Agency for Science and Innovation. Researchers from 3 NASU scientific institutions took part in it.

In the course of implementing the Law of Ukraine On Scientific and Technical Activities, several regulations were developed and adopted on state key laboratories, namely the Procedure for Granting the Status of a

[35] Resolution No.1002 of the Cabinet of Ministers of Ukraine On Approval of the State Target Scientific and Technical Programme for Research in the Antarctic for 2011-2023 of 3 November 2010. URL: https://zakon.rada.gov.ua/laws/show/1002-2010-%D0%BF#Text

[36] Access to other polar stations, ships, databases and attracting additional funds to the Antarctic program—NASC and 49 other polar organizations won the project in Horizon Europe. National Antarctic Scientific Centre. 27 July 2023. URL: *https://cutt.ly/1wYxL27e*.

State Key Laboratory in a Relevant Scientific Research and Scientific and Technical Development Area, the Model Regulation on a State Key Laboratory, model regulations on scientific and supervisory boards of state key laboratories[37] and the aforementioned CMU resolution[38] allowing for the allocation of NRFU grants to support the legally defined format of state key laboratory activities. After standardization of this form of cooperation between scientific institutions and organisations in 2023, the idea of key laboratories has not yet been developed further.

Funds for the operation and maintenance of objects of national heritage are provided for in the relevant budget programmes for the organisations managing these objects (MESU, NASU, NAASU, NAMSU, among others). The procedure for resolving issues related to the maintenance, preservation, and development of such objects, as well as the maintenance of their state register, are regulated by Order No.1233 of MESU of 13 November 2018. At the same time, no regulatory or administrative document defines the procedure for access to research or provision of services using objects of national heritage.

Ukraine is developing **electronic research infrastructures** to support scientific research and innovation.

One of its elements is the National Repository of Academic Texts[39], a nationwide distributed electronic database that accumulates, stores, and systemises academic texts. The owner of the National Repository is the State, represented by MESU, which determines the manager of the National Repository and approves the rules of procedure. In July 2018, the following regulations were approved[40].

Ukraine has established the foundations of the NASU cloud infrastructure providing resources based on the IaaS (Infrastructure as a Service) principle. This allows for efficient use of computing resources by implementing the "provision of resources on demand" model. The first cloud cluster of the Bogolyubov Institute for Theoretical Physics of NASU was built, certified, and connected to EGI Federated Cloud. This cluster is the only cloud resource in Ukraine included in the European Cloud Infrastructure and supported by EOSC. In 2018, Ukrainian National Grid (UNG) and EGI Foundation signed an associate membership agreement. Cooperation with EGI.eu provides an opportunity to build the EOSC National Initiative[41].

Since 2022, the National Electronic Research and Information System (URIS) has been under construction at the initiative of MESU, developed by the State Scientific and Technical Library of Ukraine[42]. Pursuant to the approved regulations, the system functions and evolves for collection, formation, storage, verification, provision of access to, and usage of data and information regarding scientific and scientific and technical activities, and for automation of procedures and processes of interaction between subjects and users of the system[43]. The system itself aims to solve the problem of combining all data on the results of

[37] Resolution of the Cabinet of Ministers of Ukraine On Matters of State Key Laboratories of 10 July 2019. URL: *https://zakon.rada.gov.ua/laws/show/607-2019-%D0%BF#Text*.
[38] On amendments to several Resolutions of the Cabinet of Ministers of Ukraine on the National Research Foundation of Ukraine. Resolution No.742 of the Cabinet of Ministers of Ukraine of 4 July 2023. URL: *https://zakon.rada.gov.ua/laws/show/742-2023-n#Text*.
[39] National Repository of Academic Texts. URL: *https://nrat.ukrintei.ua*.
[40] Order No.707 of MESU On Approval of the Regulations of the National Repository of Academic Texts of 04 July 2018. URL: *https://mon.gov.ua/storage/app/uploads/public/5be/54b/78c/5be54b78c6695106109232.pdf*.
[41] European open science cloud as a global instrument of scientific research. Interview of academician Anatoly Zagorodny, vice president of NASU, for Svit newspaper. 23 July 2020. URL: *https://www.nas.gov.ua/UA/Messages/Pages/View.aspx?MessageID=6661*.
[42] National Electronic Scientific and Information System (URIS). URL: *https://mon.gov.ua/ua/osvita/cifrova-osvita/elektronna-naukovo-informacijna-sistema*.
[43] Resolution No.1067 of the Cabinet of Ministers of Ukraine On Approval of the Provision on the National Electronic Scientific and Information System of 27 September 2022. URL: *https://zakon.rada.gov.ua/laws/show/1067-2022-%D0%BF#n121*.

professional scientific and technical activities of Ukrainian scientific institutions, higher education institutions, and scientists. The latter use various information services during research to process information or store research data, and sometimes information about research activity is either not preserved (or not available at all) or stored in different databases, which consequently makes it difficult to find and reuse. Currently, the system has the following functions:

- aggregation of data on the sphere of scientific and technical activities of Ukraine;
- automation of certain procedures and processes of the subjects of the Ukrainian science sector;
- ensuring generation of data which are not available in accessible information sources;
- ensuring confirmation of data by verified System users;
- combining data, their presentation, and provision of analytical tools to users for their research;
- aggregation of information useful for domestic scientists, employees of scientific institutions and higher education institutions, on a single resource.

Performance of this system's functions is aimed at ensuring the openness of data on Ukrainian science and simplifying the process of finding information, data, equipment, services, and resources for research and procedures for filling out applications and reports for Ukrainian scientists, research institutions, and higher education institutions. Since September 2023, MESU has used the system to conduct two traditional tenders for scientific R&D projects. In the first competition alone held through the nauka.gov.ua portal[44], nearly 750 research teams from Ukrainian higher education institutions used the new functionality to submit grant applications.

In the context of various risks, the storage, management, and exchange of research data (data obtained from research or RI-related data) on the basis of Open Science and FAIR principles (Findability, Accessibility, Interoperability, Reusability) are assigned a special role. Meanwhile, communication during the focus group with MESU CSUSE and NASU representatives demonstrates these practices in the Ukrainian RI are literally absent.

2.3 The state of Ukrainian research infrastructure

The modern RI system in Ukraine is characterized by outdated and underfinanced equipment and property, a lot of which has suffered damages or even destruction since the Russian invasion.

Ukraine's RI system is a legacy of the extensive Soviet research infrastructure, which was primarily focused on industrialising the country and, in line with the cooperation and industrial specialisation of the time, was

[44] 750 projects were submitted to the scientific R&D project tender held by MESU. URL: *https://nauka.gov.ua/news/na-konkurs-proiektiv-naukovykh-doslidzhen-i-rozrobok-mon-podano-750-proiektiv/*.

erected in close proximity to major industrial centres to facilitate their growth and support. This had some positive consequences as the established economic clusters ensured industrial development in certain regions of the country.

Since 1991, Ukraine has lost most of its economic sectoral linkages and begun to plan expenditures based on available resources and new economic challenges. The scientific system, and RI as part of it, failed to timely respond to technological developments that involved greater digitalisation and re-orientation of key industries towards new materials, technologies, and new customers. Most of Ukraine's RI failed to re-orient to the new conditions, which led to a reduction of investments and customers' focus on off-the-shelf technologies outside Ukraine.

The Final Report of the Independent European Audit of the National Research and Innovation System of Ukraine conducted in 2016 noted that "Ukraine inherited about 20% of the Soviet Union's experimental centres, including nuclear reactors, astronomical observatories and marine research vessels; however, much of this infrastructure was lost during the period of independence"[45].

Lack of coordination and strategic planning in science and research constitutes a major problem. It was primarily caused by government policies that reduced funding for science in general and RI in particular. Additionally, the political and economic turbulence in Ukraine has affected the opportunities for RI development. Consequently, the research infrastructure is outdated for the most part, which limits the possibility of conducting research on it. Many research institutions and universities had insufficient funds to conduct research at a level that would ensure their competitiveness. As a result, highly qualified researchers, including young people, are often forced to seek opportunities abroad.

The reasons for the problems in the functioning of the Ukrainian RI are summarised in the Concept of the State Target Programme for the Development of Research Infrastructures in Ukraine until 2026[46]:

- long-term underfinancing of capital expenditures for maintenance and renewal of the material and technical base of research institutions and higher education institutions;
- insufficient extra-budgetary receivables for RI, in particular, revenues from the use of scientific equipment;
- high degree of wear of scientific equipment in scientific institutions and higher education institutions;
- lack of flexible instruments for redistribution and accumulation of funding for upgrading the material and technical base of research institutions, higher education institutions, and RI, specifically, opportunities to place temporarily free budget funds received for the provision of paid services by research institutions, higher education institutions, and research institutes, on deposit accounts in public sector banks;
- lack of systematic development of RI and state policy in this area;
- imperfection of the regulatory framework for the establishment and development of RI;
- lack of a procedure for identifying and monitoring Ukrainian research infrastructures, as well as systematisation and digitalisation of relevant data, particularly the absence of a RI system database.

[45] European Commission, Directorate-General for Research and Innovation, Schuch, K., Kiopa, A., Posselt, T. (2017). Final Report of an Independent European Audit of Ukraine's National Research and Innovation System. Horizon 2020 Policy Support Facility. URL: *https://data.europa.eu/doi/10.2777/462461*

[46] Розпорядження Кабінету Міністрів України від 14.04.2021 № 322-р "Про схвалення Концепції Державної цільової програми розвитку дослідницьких інфраструктур в Україні на період до 2026 року". URL: *https://zakon.rada.gov.ua/laws/show/322-2021-p#Text*.

Among the reasons stated in the Concept, the most frequently mentioned in expert interviews and focus groups were the long-term underfunding of capital expenditures for the RI upgrade and the high level of wear of scientific equipment. Another important problem is the funding of salaries for the technical staff maintaining research facilities.

For example, high-performance computing requires hardware upgrades every 3–5 years; it is also important to raise the status of the computer system administrator, who is not a scientist in the scientific hierarchy, with the corresponding level of salary which is several times lower than in private commercial IT companies. The approach of the cloud architecture also needs to be revised from "infrastructure as a service" (IaaS) to "platform as a service" (PaaS), which would optimise costs and increase efficiency.

Ukraine has not yet implemented the practice of RI mapping. Mapping should be viewed not only as an inventory taking (which is often available on the websites of scientific academies or universities and is currently planned for URIS) but also as a process of prioritising strategically important and long-term funding for RI based on the efficiency of its use.

Thus, one may conclude that there is no nationwide network of research infrastructures in Ukraine, and there are problems with information on the set of equipment and facilities, collections, archives, etc., no common rules for their use, no coordinated management and coordination system, and no special funding instruments. At the same time, there is no practice of ex ante monitoring and prioritisation of funding for RI based on its potential capacity.

The RI governance system is fragmented, with an extensive classification of RI types, but the semantics are not meaningful, such as the distribution of certain functions or capabilities between different types of infrastructure, and it is more like different names than differentiated models for managing certain types of RI. One example of a RI, the CSUSE, is discussed as a separate case study in this section of the paper.

Collaboration between science and business in the RI development (resource and expertise aspect) could potentially solve some of these issues. However, due to the lack of effective public-private partnerships, it is not currently a factor.

Furthermore, RIs are not part of regional innovation systems and are not an asset for the development of smart specialisation with relevant technologies, needs, investments, business demands, and human capital provision.

Impact of war

The systemic problems in the research infrastructure area have been significantly aggravated by the Russian aggression since 2014, when the RI of the Autonomous Republic of Crimea and partially the one of Donetsk and Luhansk regions were lost, and even more by the full-scale invasion in 2022, when RI equipment in many cities of Ukraine was destroyed, damaged, or removed by the invaders. By March 2023, 35% of Ukraine's research infrastructure was damaged or destroyed, compared to 15% in November 2022[47].

The destruction of RI continues due to constant shelling; therefore, information on the destruction is being collected, a database of damaged and destroyed research infrastructure[48] is being formed, and information on the loss calculation methodology is being

[47] Council of the European Union (2023, June 2). ERAC Report on the role of Research and Innovation in Ukraine's Recovery. URL: *https://www.parlament.gv.at/dokument/XXVII/EU/143295/imfname_11256107.pdf*.

[48] Database of damaged and destroyed research infrastructure. URL: *https://scienceatrisk.org/uk/whitepaper/bazy-danykh-poshkodzhenoi-ta-zruinovanoi-doslidnytskoi-infrastruktury*.

systematised. The RI that suffered the most are located in the regions close to the combat zones, in particular Kharkiv and Kyiv, as well as in the territories that have been liberated from occupation.

For instance, as of September 2022, dozens of buildings had been destroyed and damaged in the institutions of the Northeast Science Centre alone, and 70%t of national heritage facilities were damaged. This includes, inter alia, the Neutron Source nuclear subcritical facility of the National Science Centre "Kharkiv Institute of Physics and Technology" of NASU; literature monuments collection of the Central Scientific Library of V. N. Karazin Kharkiv National University; the plant genetic resources bank of the Plant Production Institute named after V. Ya. Yuriev of NASU, and more. As a result of the shelling, the radio astronomy observatory was severely damaged, including the world's largest decametre wavelength radio telescope, UTR-2, and the GURT telescope. Many scientific facilities have been mined[49].

During the interviews conducted for this study HEIs and SIs representatives that were displaced in 2014–2015 from the territories occupied by Russia (Vasyl' Stus Donetsk National University, Institute of Applied Mathematics and Mechanics of the National Academy of Sciences of Ukraine, State Organisation "V. Mamutov Institute of Economic and Legal Research of the National Academy of Sciences of Ukraine"), it was noted that no proper conditions had been created for the evacuation of scientific equipment. The process of equipment evacuation was also complicated by the introduction of thorough inspections at checkpoints set up by the militants of the so-called "DPR" (Donetsk People's Republic) and the pro-Russian position of some colleagues who prevented employees from entering the premises and, accordingly, taking out any equipment or documentation. Nevertheless, some computer equipment and electronic infrastructure were successfully evacuated, but this was accompanied by personal risk for the scientists who were literally tasked with preserving the integrity of the scientific infrastructure. What's more, university and NGO bank accounts were blocked for almost six months during 2014, forcing academics to fund relocations out of their own pockets. HEI and SI CSUSE representatives in northeastern Ukraine who have been victims of artillery shelling and bombing since the beginning of the full-scale invasion of the Russian Federation in February 2022 until the time of this writing, expressed concerns about the lack of a clear algorithm for the scientific equipment evacuation.

Funding

The following case study demonstrates the differences in the funding of RI through public funds and international support and illustrates the gap in this area in Ukraine.

In 2023, the University of Tartu (UT) received the largest funding in the history of Estonia for research in bioindustry digitalisation and personalised medicine. Thus, to establish two international research centres — The Centre for Digitalised Bioengineering and the Research and Development Centre for Personalised Medicine — UT received EUR 30 million for each centre from the European Commission and another EUR 30 million from the Government of Estonia[50]. It should be noted that this amount of funding for these two projects is equivalent to UAH 3.6 billion and corresponds to the four-year expenditure of the budget programme under KPKVK 2201040 (Programme Classification of Expenditures and Local Budget Lending) which finances research and development in Ukrainian HEIs. Considering this, it is difficult to expect Ukrainian scientists to be competitive

[49] A scientific centre on the firing line. Svit. 2022. ##35–36. URL: *https://svit.kpi.ua/2022/09/19/науковий-центр-на-лінії-вогню/*.

[50] Punamäe, O.M. (2023). University of Tartu scientists received the largest research grant in history. Eesti Rahvusringhääling. URL: *https://rus.err.ee/1608843334/uchenye-tartuskogo-universiteta-poluchili-krupnejshij-v-istorii-jestonii-grant-na-issledovanija*.

with their European counterparts. Moreover, research and development in Estonia received EUR 386 million from the budget in 2023[51]*.*

Funding for science is growing both at universities and enterprises — since 2020, the manufacturing industry which accounts for the major share of Estonian exports has increased its spending on science the most. Furthermore, since 2020 the number of scientists and engineers employed in the private sector has been growing, with the number of full-time employees increasing by 220 in 2021 alone. And since 2018, the number of scientists and engineers at enterprises has doubled. These changes are expected to have an impact on labour productivity in Estonia. Since 2010, the productivity of the Estonian labour force has increased by 17% compared to the EU average and reached 86.7% of the EU average in 2021. This creates the preconditions for achieving the target of 110% of the European average by 2035.

Following the competition held by MESU in 2022, 40 projects were selected (out of 195 submitted proposals, of which 156 met the formal requirements of the competition and were submitted for scientific examination) in four areas of funding for a total amount of UAH 114,605.43 thousand (budget programme under KPKVK 2201380). At the same time, only six out of 36 applications that were recognised as winners of the competition were infrastructure projects, with a total expenditure of UAH 48.42 million[52]. Simultaneously, the only university that, according to an interview with a Horizon 2020 international INFRA project participant, could have joined the pan-European RI on the condition of purchasing special equipment with national funding, failed to receive support from MESU.

The HEIs are already trying to raise funds for reconstruction from foreign sources. When communicating with foreign donors, some respondents receive feedback that no investments will be made in the RI until the end of hostilities.

The respondents also note the low involvement of research and teaching staff in initiating international RI development projects due to low awareness of them and, in fact, the lack of necessary logistic means and thus the inability to offer new methods, approaches, and elements that can add value and complement the existing international consortia.

A participant in the international ACE-EGI project of the EU's Horizon 2020 framework programme has expressed doubts about the feasibility of future cooperation with consortium partners and the ability to submit future applications for respective competitions due to a lack of powerful equipment or programmers suitable for tasks solved at the EU project level. Olena Orliuk, the Director of UANIPIO, said in an interview there was a problem of unreliable results due to scientists using uncertified equipment for their work.

The amount of funding received by Ukrainian organizations through the competitive funding of projects within the European Union's Horizon 2020 Framework Programme for Research and Innovation was EUR 44.92 million. A total of 240 grant contracts have been signed under this programme, including only 14 for a total of EUR 469.37 thousand under the Research Infrastructures subject area. Among the 15 associated countries, Ukraine ranks seventh in terms of the number of successful

[51] State budget of Estonia allocated EUR 386 million for research and development activities in 2023. Eesti Rahvusringhääling. 24.01.2023. URL: *https://rus.err.ee/1608860666/na-nauchnye-issledovanija-i-opytno-konstruktorskie-raboty-v-jetom-godu-vydeljat-386-mln-evro*.

[52] 40 Ukrainian projects will receive financial support from the European Union's Horizon 2020 Framework Programme for Research and Innovations. MESU. URL: *https://mon.gov.ua/ua/news/do-ramkovoyi-programi-yevropejskogo-soyuzu-z-naukovih-doslidzhen-ta-innovacij-gorizont-2020-vidibrano-40-ukrayinskih-proyektiv*.

projects and the amount of funding received from the EU[53].

Comparative data on the participation of the Eastern Partnership countries in projects to establish and develop research infrastructures under the Horizon 2020 programme are presented in Table 2

Table 2. Number of projects submitted for INFRA tenders in the Eastern Partnership countries

Country	Total number of projects	Number of RI projects, (%)
Azerbaijan	20	4 (20%)
Belarus	55	6 (10%)
Armenia	37	7 (21%)
Georgia	58	8 (14%)
Moldova	73	11 (15%)
Ukraine	240	12 (5%)

As Table 2 shows, despite the largest total number of projects involving Ukrainian teams, participation in research infrastructure projects accounts for the smallest share among the countries listed. It is worth noting that Azerbaijan and Belarus were not Horizon 2020 associated countries, so the participation of these teams in such projects was rather a recognition of their capacity and importance for the European Research Area. The development of RI is fundamentally important for ensuring high-level research in general and requires sufficient government involvement in the national RI development, so the number of projects correlates with the availability of RI development tools and financial support for the development of the scientific sphere in each of the Eastern Partnership countries. Obviously, compared to Armenia, Georgia, and Belarus, the RI development level in Ukraine is insufficient for active participation and visible partnership in the European Research Area. As of 28 August 2023, Ukrainian teams participate only in two projects under the Horizon Europe INFRA competitions.

Limited domestic resources and research opportunities require scientists to use new strategies and tools. To obtain competitive research results, an increasing number of Ukrainian scientists use international mobility programmes. This trend has become particularly prominent during the war, as many foreign universities and research institutions have initiated programmes to support researchers, including those who have left Ukraine. However, under martial law, men face travel restrictions, which creates difficulties in obtaining permits for temporary border crossing. This raises the issue of developing hybrid and remote formats for accessing RI.

RI is critically essential for the development of science, but society does not perceive problems in this area. And yet, in the face of national security threats that require the involvement of science, these problems become visible to society. This was the case in 2020, when it turned out that Ukraine's development of its own vaccines, specifically to combat the coronavirus, was impossible due to the lack of the necessary RI.

The production of modern vaccines is a high-tech, knowledge-intensive business that requires

[53] R&I country profile. URL: *https://dashboard.tech.ec.europa.eu/qs_digit_dashboard_mt/public/sense/app/1213b8cd-3ebe-4730-b0f5-fa4e326df2e2/sheet/d23bba31-e385-4cc0-975e-a67059972142/state/analysis*.

significant RI investment over time. Moderna Therapeutics, the U.S. company that announced the release of their vaccine in September 2020, received funding in the amount of USD 240 million from the Swedish-British pharmaceutical giant AstraZeneca in 2013, and another USD 100 million from the US-based Alexion in 2014. Since 2015, it has received USD 300 million from Merck, one of the world's largest pharmaceutical companies. To enable Moderna to produce a coronavirus vaccine within a short time in 2020, it has received hundreds of millions of investments over the past decade, developing its scientific and industrial capacities and engaging the best specialists. Funding for NASU is less than USD 200 million a year, and for the NAMSU it's even less, USD 50 million. That is, the entire state-funded science of Ukraine is losing out on funding to one American company[54]. According to NASU academician Serhii Komisarenko, Ukrainian scientists can create a vaccine against coronavirus, but Ukraine does not have the technical capabilities for mass production. The scientist noted that Ukraine does not possess a genome centre, the right to work with viruses, BSL-3 reference laboratories, or a biotechnology centre that would scale up what is produced in the laboratory for mass production[55].

2.4 Best practices for research infrastructure development at national level

The overall strategy for the establishment and development of RI is reflected in national roadmaps and action plans developed with the involvement of experts from various scientific fields, economists, and other stakeholders and approved by relevant government decisions. As of today, roadmaps of 27 EU countries[56] and associated countries are available for analysis.

RI development roadmaps are critical policy documents for strategic planning. They serve as a reference point for planning the future development of science, providing a comprehensive overview of existing RI capabilities, and also provide decision-makers, researchers, and businesses with an understanding of the current state of RI for more effective decision-making at all levels. Most importantly, roadmaps help policymakers to attract investment in RI more effectively in line with its domestic and international significance. In addition, roadmaps give the business community a better understanding of the country's research potential. This is a key link to establishing a closer cooperation with academia. Finally, RI development roadmaps can also engage a broader audience and improve visibility and recognition of domestic research.

The history of RI roadmaps created by the EU and associated countries that have undergone political transformation processes, including in the context of military conflicts, suggests that the allocation of RI to separate structural elements of science should be preceded by the achievement of a certain development level of the country's entire scientific, technological, and innovation system. A remarkable factor here is the access of countries to structural

[54] Feia, O. (2020). Half a kingdom for a vaccine. Tyzhden, 17 April 2020. URL: *https://tyzhden.ua/pivtsarstva-za-vaktsynu/*.
[55] Academician Serhii Komisarenko's comment: "Ukraine cannot produce COVID vaccines." 16 October.2020. URL: *https://www.nas.gov.ua/UA/Messages/Pages/View.aspx?MessageID=7054*.
[56] National Roadmaps. URL: *https://www.esfri.eu/national-roadmaps*.

funds, regional development funds or other EU programmes.

The Czech Republic's first RI roadmap was developed and approved by the government in 2010, i.e. six years after the country's accession to the EU, and Slovakia's only in 2020[57]. Each of them envisages the provision of resources for the establishment of a new RI from national budgets, available EU funds, and also through joint projects with existing RIs of pan-European importance that receive funding from EU framework programmes. Each project has a deadline after which its results are evaluated, and a decision is made on the feasibility and possibility of its continuation. The Czech Roadmap includes RIs that are not included in the ESFRI plan (such as transport) but are of great importance for the Czech Republic, are linked to RIs of other countries (Japan), and in which, according to the country's needs, major state budget funds have been invested. One of the main selection criteria is the uniqueness of the specific RI. In other words, if there are similar RIs in the country, none will be funded, or they join forces and act as one RI distributed by location.

The development strategy for the Czech Ri (Evaluation Methodology for Research Infrastructures (2014)[58] includes the following aspects:

- sustainability and development, including a short-term budget horizon as well as a long-term perspective based on the service life of key equipment (for instance, 2–3 years for ICT or 5–10 years for standard equipment), which in turn should be reflected in the depreciation charges for the renewal of such equipment;
- staff development (employment strategy, defined career procedures (rules) aimed at professional development of employees);
- communications (internationalisation and promotion, PR strategy);
- internal strategic research (research aimed at improving services for users, research to develop the infrastructure's capacity, support for user research, joint and contract research);
- usage at national or transnational level (a clearly articulated and transparent strategy for providing access to RI to different user groups, defining the manner and conditions of cooperation with other research organisations, defining open access mechanisms ("open access" and "transnational access"), analysing capacity and funding based on the scientific merit of the proposal, addressing ethical, security, and misuse issues).

The Roadmap of Slovakia provides for RI funding from the state budget (state programmes, targeted programmes and projects, self-sufficiency through service provision), private businesses, and EU structural and operational funds which make up a significant share in the research sector funding in Slovakia[59].

As for the Western Balkan countries, only three of them have RI roadmaps, namely Slovenia (2011, updated in 2016 and 2020), Croatia (2014), and Montenegro (2015, updated in 2019 in line with the later developed smart specialisation strategy), while Serbia, North Macedonia, Bosnia and Herzegovina, and Albania are at different stages of developing similar documents. It should be noted that roadmap development processes are lengthy and involve preliminary consultations with a wide range of stakeholders. The expediency of developing a common regional policy for access to all RIs for researchers from this region and creating a regional roadmap with the support of the Regional Cooperation Council[60] is

[57] National Roadmaps. URL: *https://www.esfri.eu/national-roadmaps*.
[58] Evaluation Methodology for Research Infrastructures (2014). URL: *https://www.msmt.cz/file/33846_1_1/*.
[59] Roadmap of Research Infrastructures – SK VI Roadmap 2020–2030. URL: *https://www.minedu.sk/data/files/11490_roadmap-of-research-infrastructures-sk-vi-roadmap-2020-2030-eng.pdf*.
[60] Western Balkans agree to step up the work on research infrastructure. URL: *https://www.gov.me/en/article/193172--western-balkans-agree-to-step-up-the-work-on-research-infrastructure*.

considered after the completion of national processes for the development of RI.

Establishment of new RIs, upgrading the existing ones, or joining international ones requires additional budget investments. This is why data-driven decisions are necessary to increase the efficiency of public funds. Possible mergers and concentrations of efforts are being considered for the same purpose.

During the intermediate stages of implementation and enforcement, RIs provide access to the use of their resources on a competitive basis to researchers from research institutions of all forms of ownership and universities, as well as industrial/commercial organisations. Such competitions are periodically announced by the EU Joint Research Centre (JRC). The RICH Europe project establishing a network of NPCs for the development of RI provides constant information on the possibilities of using RI.

The main strategic areas for the development of a pan-European RI in the updated ERA are formulated in work programmes which are updated during the implementation of the framework programme based on ESFRI recommendations and monitoring of the activities of those RIs that have passed all development phases from RI concept design in general to operational ones, i.e. conducting advanced research, providing services to the scientific community, improving methods and tools, and obtaining political and financial support.

It is worth noting that ESFRI offers quite detailed indicators for RI monitoring which consider promotion of scientific excellence, educational aspects of the RI functioning, international cooperation, cooperation with business, public relations and popularisation of science, optimisation of data management, consulting activities, and a business model[61]. Digitalisation of operational processes and use of RI[62] through the Internet of Things would enable automatic online monitoring and automated transparent reports. In the future, a more comprehensive list of monitoring indicators for RI could be used, taking into account sustainable development.

Some Ukrainian HEIs/SIs research infrastructure has already shown readiness to implement monitoring indicators as they are included in international rankings, some of which take RI into account. One of the best practices identified during the focus group is the procedure developed by Sumy State University for consideration of environmental impacts in scientific, research, and innovation activities and the implementation of their results. According to the procedure, consideration of environmental consequences in the course of scientific research is mandatory in cases where laboratory, analytical, technological, and other specialised equipment is used, the use of which causes environmental impacts subject to individual regulation pursuant to the requirements of the environmental legislation of Ukraine (generation of waste, wastewater, emissions into the atmosphere).

For RI monitoring purposes, it will also be useful to draw on the long-term experience of the U.S. Office of Scientific and Technical Information (OSTI) in collecting data on R&D results or scientific and technical information from laboratories and presenting them to a wide audience through an information portal[63].

One must consider best practices for the RI management at the operational level. Research infrastructure management systems (RIMS) have significant potential in this regard. The implementation of RIMS facilitates the

[61] ESFRI (2019). Monitoring of Research Infrastructures Performance. Working group report. URL: *https://www.esfri.eu/sites/default/files/ESFRI_WG_Monitoring_Report.pdf*.
[62] A practical guide on monitoring research infrastructures. 2017. URL: *https://www.interreg-danube.eu/uploads/media/approved_project_output/0001/32/8944de08315bc53e8f659fb6e131f2d602279905.pdf*.
[63] DOE National Laboratories, Facilities, and Programs. URL: *https://www.osti.gov/doe-national-laboratories*.

identification and availability of research infrastructure for researchers and external users from the sector and industry. This enables them to obtain the best research results by using the most appropriate equipment, collaborating with leading researchers, making the most of research equipment, and engaging in mutually beneficial interactions with industry through better procurement.

Cooperation at national and international levels is crucial. For instance, the WISE case[64] is based on the collaboration of seven prominent research universities in Sweden. Together, they form a complete and complementary knowledge base for successful research and education. The long-term objective of WISE is to facilitate and enhance the transition to a sustainable society, while establishing Sweden as a leading country in materials science. An essential aspect of this work is to establish national infrastructures with research facilities that facilitate innovative research of the highest quality.

In general, best global practices show that public-private partnerships are common in the development of RI (RI administration, co-financing, providing access to private RI for researchers, and cooperative models within smart city projects). However, initial investments in RI development were made by the State or structural and other funds available in a given country.

Several important conclusions and recommendations for Ukraine in restoring and supporting RI can be drawn from these cases and examples: the need for comprehensive support programmes that include not only the allocation of funds, but also the development of clear key performance indicators (KPIs) and linkage to the real needs of the country's post-war recovery in the short and medium term; building models of cooperation at the national and international levels to ensure access to RI and implement initiatives for its development.

2.5 CSUSE case study

Centres for the Shared Use of Scientific Equipment (CSUSE) are one of the organisational and legal forms of RI for conducting research and developing science in Ukraine. This organisational form of RI operation is the most widespread and is focused on providing opportunities for research groups from different fields of science to jointly use specialised equipment that is expensive, difficult to manufacture, or requiring specialised maintenance. Since the CSUSEs are established using the facilities of scientific institutions and higher education institutions, the network is divided into CSUSEs managed by different key budget managers — the Ministry of Education and Science of Ukraine and the National Academy of Sciences of Ukraine.

CSUSEs provide access and the right to use devices and equipment free of charge to employees of a scientific institution, university, academy, or institute within which the centres are established, and other persons performing research work, in accordance with the conditions specified in the regulations on access to, and use of, scientific equipment. The Model Regulation on CSUSEs was approved by CMU Resolution No.703[65] of 21 June 2022. Prior to that, the CSUSEs operated pursuant to their own regulations.

[64] Wallenberg Initiative Materials Science for Sustainability. URL: *https://wise-materials.org/*.
[65] Resolution No.703 of the Cabinet of Ministers of Ukraine On Several Matters of Centres for Shared Use of Scientific Equipment of 21 June 2022. URL: *https://zakon.rada.gov.ua/laws/show/703-2022-%D0%BF#Text*.

2.5.1 CSUSE of MESU

In 2018–2019, 21 CSUSEs[66] were established in higher education institutions under the management of MESU. Their establishment was aimed at strengthening the material and technical laboratory base in HEIs in all regions of Ukraine. Only five out of 21 centres were opened in Kyiv (using the facilities of National University of Food Technologies, National University of Life and Environmental Sciences of Ukraine, Mykhailo Drahomanov Ukrainian State University, National Technical University of Ukraine "Igor Sikorsky Kyiv Polytechnic Institute" and Taras Shevchenko National University of Kyiv). Scientists from the following state-owned HEIs were given the opportunity to develop research infrastructure: Uzhhorod National University (Uzhhorod), Pryazovsky State Technical University (Mariupol), Dnipro University of Technology and Oles Honchar Dnipro National University (Dnipro), Ivan Franko National University of Lviv and Lviv Polytechnic National University (Lviv), Sumy State University and Sumy National Agrarian University (Sumy), Kharkiv National University of Radio Electronics, V. N. Karazin Kharkiv National University and National Aerospace University "Kharkiv Aviation Institute" (Kharkiv), Odesa I. I. Mechnikov National University (Odesa), Ivano-Frankivsk National Technical University of Oil and Gas and Vasyl Stefanyk Precarpathian National University (Ivano-Frankivsk), Poltava State Agrarian Academy (Poltava), National University of Water and Environmental Engineering (Rivne). The University Consortium CSUSE, founded by the State Higher Educational Institution "Pryazovsky State Technical University" (Mariupol), is located in the temporarily occupied territory of Ukraine and its fate remains unknown.

The Concept of the State Target Programme for the Development of Research Infrastructures in Ukraine until 2026, approved in 2021, envisaging the establishment and operation of at least 50 centres for the shared use of equipment[67]. However, achievement of this figure may be unrealistic in the current and post-war conditions. Funding for the activities of CSUSEs under the management of MESU should be provided by the Budget Programme 2201040 "Scientific and Technical Activities of Higher Education Institutions and Research Institutions". From 2020 onwards, expenditures for the purchase of high-tech research equipment will not be financed from the general fund (in 2018, UAH 31,872.0 thousand was financed, and in 2019, UAH 32,955.7 thousand was financed at the expense of the general fund)[68]. Therefore, the development and upkeep of the CSUSEs even without state funding should be guaranteed through a dedicated fund, such as the HEI's own revenues, which can be generated by providing scientific services, securing international grants, and other means.

[66] On Establishment of Centres for Shared Use of Scientific Equipment, MESU Orders No.1439 of 19 November 2019, No.475 of 12 April 2019, and No.444 of 02 May 2018.
[67] Order No.322-p of the Cabinet of Ministers of Ukraine On Approval of the Concept of the State Target Programme for the Development of Research Infrastructures in Ukraine until 2026 of 14 April 2021. URL: *https://zakon.rada.gov.ua/laws/show/322-2021-%D1%80#Text*.
[68] Reports on the execution of budget programme passports in 2018. MESU. URL: *https://mon.gov.ua/storage/app/media/budzhet/2019/03/pasporta-za-2018.pdf*; *https://mon.gov.ua/storage/app/media/budzhet/2020/07/15/Zvit%20byudzhetnykh%20prohram%20za%202019.rar*

2.5.2 CSUSE of NASU

In the early 2000s, the reorganisation of the system of use of scientific equipment owned by individual research institutions was identified as one of the ways to improve the efficiency of NASU scientists and create a competitive scientific product.

It should be noted that during the first 10 years of Ukraine's independence, there was virtually no renewal of scientific instruments and equipment, primarily those imported, due to lack of funds. The amount of funds allocated from the State Budget of Ukraine failed to meet even the minimum requirements that would enable maintaining the proper level of experimental research, while the dedicated funds of individual institutions were, and still are, extremely insufficient for the purchase of unique and expensive equipment for modern experimental research. Depreciation charges on such equipment were not made due to ambiguity of interpretation, which requires an audit of accounting principles and standardisation of approaches and processes across all research institutions. At the end of 2003, the Law of Ukraine *On the State Budget of Ukraine* for 2004 included Article 104, which allowed NASU to purchase imported scientific equipment from direct manufacturers in 2004 without a tender procedure, according to the list approved by the Cabinet of Ministers[69]. This was the beginning of the establishment of the CSUSE of NASU that lasted almost five years; no funds were allocated from the general fund of the state budget for these programmes thereafter.

The establishment and further operation of the CSUSE at NASU is regulated by Order No.322 of the NASU Presidium of 28 April 2004, which provides for the right to use equipment free of charge for NASU institutions and, under joint activity agreements, for higher education institutions, and under service agreements for third-party organisations. Since 2013, foreign scientific institutions have also become consumers of such services under joint research agreements. The dynamics of the acquisition of scientific equipment and the establishment of the CSUSE is shown in Fig. 1. The aggression of the Russian Federation against Ukraine caused losses to the CSUSE as early as 2014–2015. In 2014, three CSUSEs ceased their operation as a result of the annexation of the Autonomous Republic of Crimea, and in 2015, two CSUSEs were lost as a result of the occupation of Donetsk. As part of the optimisation of resources, the CSUSEs located in one institution were merged in 2016.

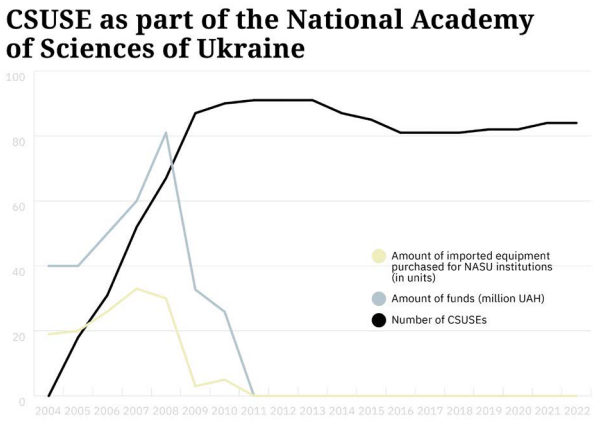


Figure 1. Development of CSUSEs at research institutions of MESU[70]

Since a major part of the CSUSE equipment has been in operation for more than 15 years, some of the equipment that continues to be produced by foreign companies in the scientific and instrument-making industry has already undergone significant modernisation. At the same time, there is a critical shortage of funds to upgrade the existing equipment, while the cost of repairs has a steady upward trend.

[69] Report on the activities of the National Academy of Sciences of Ukraine in 2003. Part 2. Kyiv, 2004.
[70] Report on the activities of the National Academy of Sciences of Ukraine, Kyiv, Akademperiodika, 2004, 2005....2022.

Given the need to support the sustainable functioning of CSUSEs, a programme was launched in 2009 to provide them with centralised supply of consumables and repair out-of-service units and components of instruments and equipment. Since 2011, the general fund of the state budget has been the only meaningful source of funding for CSUSEs despite annual revenues from the provision of services to third-party organisations.

The dynamics of the use of funds for the purchase of equipment and instruments for the NASU institutions is shown in Fig. 2. It is worth noting that the share of funds from the relevant budget programme in the total NASU funding under this item of expenditure in 2009-2011 dropped significantly due to the global financial crisis and the devaluation of the currency exchange rate against the hryvnia, and in 2015 it was aggravated further by Russia's military aggression against Ukraine. This share continued to decline after 2015.

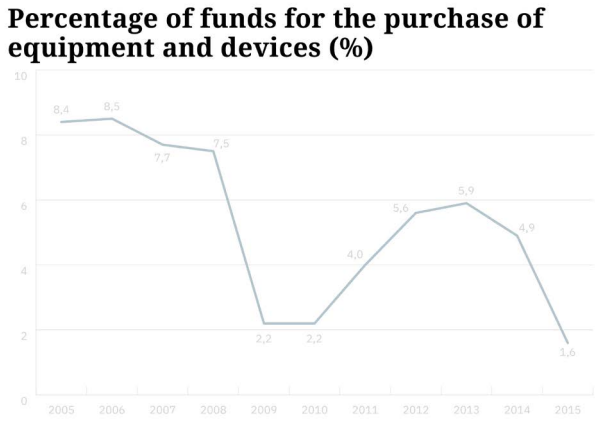


Figure 2. Share of funds of the total NASU funding, allocated for purchase of equipment and devices for NASU institutions in 2005–2015[71]

In 2014, UAH 23.6311 million of budget funds was spent under Budget Programme 6541030 for "technical maintenance of scientific facilities of the National Academy of Sciences of Ukraine", and UAH 26.7114 million was spent on "ensuring preservation and proper functioning of scientific facilities of the National Academy of Sciences of Ukraine that have the status of national heritage". In the next two years, the amount of funds for these activities decreased to UAH 21.715 and 25.7524 million in 2015 and to UAH 18.8 million and 22.4 million in 2016, respectively; however, from 2017 onwards, the funding for technical maintenance of the NASU scientific facilities grew once more and amounted to UAH 29.497 million and UAH 28.2 million, respectively, and in 2018—UAH 27.4 million and UAH 31.0 million.

Separate funding for the acquisition of new and modernisation of existing scientific equipment was provided by the National Academy of Sciences of Ukraine under Budget Programme 6541230 "Support for the development of priority scientific research areas", which began in 2018 as a pilot project to support research teams of institutions that demonstrated better performance based on the results of the performance assessment[72].

[71] National Academy of Sciences of Ukraine (2010). Short annual reports 2009–2015. URL: *https://files.nas.gov.ua/text/report/2009_ua.pdf*.
[72] Report on the execution of budget programme passports. NASU: *Home Page (nas.gov. ua)*

2.5.3 Problems in the functioning of the Ukrainian CSUSEs

During our meeting with experts, representatives of the seven CSUSEs established in HEIs and research institutions of the National Academy of Sciences of Ukraine, a range of problems was outlined in the functioning of such infrastructures, the solution of which should optimise their activities. These include the following:

- lack of openness of the CSUSEs and an appropriate communication strategy, including insufficient Internet coverage of information about equipment and contacts, which results in potential users (HEIs, SIs, businesses, or individuals) not being sufficiently aware of the activities and capabilities of such centres, and therefore cannot use them to the benefit of the CSUSEs. As a solution to this problem, respondents suggested creating an organisational tool that would consolidate information on access to the material and technical facilities of all the Centres, including online reservations and payment for use for individuals and legal entities;
- insufficient funding for the development and renewal of the laboratory material and technical facilities of the CSUSEs;
- lack of full-time CSUSE scientific and engineering staff, as scientists are involved in the work of the CSUSEs mainly on a voluntary basis, part-time, or through internal incentives from the HEI administration;
- complexity of calculations of the estimated cost of scientific services provided by the CSUSE (ambiguity in accounting for depreciation costs, consumables and utilities, labour costs in the absence of full-time researchers), which may be classified as a violation of budget legislation;
- providing CSUSE services on a commercial basis only (instead of free access for budgetary institution representatives), given the potential negative consequences during inspections by the State Audit Service of Ukraine;
- problems with tender procurements of unique scientific equipment due to unfair competition and the availability of "fake" equipment;
- lack of stable budget funding for the CSUSEs, and therefore their functioning is solely the responsibility of the HEI/SI;
- lack of systematic support for professional development of scientists involved in the activities of the Centres;
- lack of algorithm of actions in case of unforeseen circumstances and hostilities — how the safety and evacuation of the CSUSE scientific equipment should be ensured.

Olena Filimonova, ERASMUS+KA2 project manager at the DEFEP of the Black Sea National University named after Petro Mohyla, noted during an interview that despite the availability of regulatory and legal solutions in Ukraine allowing the development of RI, in particular the Law of Ukraine On Public Procurement and additional bylaws that provide for simplification of customs procedures and value-added taxation (VAT) in case of equipment procurement under international cooperation projects and logistical assistance, there are problems with their interpretation and practical implementation, in particular at the level of HEIs or the State Treasury Service of Ukraine.

The common problems of Ukrainian RI management also affect the functioning of the CSUSEs. The management problems of the CSUSEs' functioning at HEIs include lack of systematic monitoring of their activities by the Ministry of Education and Science of Ukraine, which hinders effective financial support for the best RIs. The results of the

CSUSEs' activities are not published in open sources, which makes it difficult to assess their impact and the importance of their very existence for scientists and the public. The scientists' need to use the equipment, the degree of workload of unique instruments, and the priority of access are not analysed by either NASU or MESU.

SECTION 3. WAYS OF RESEARCH INFRASTRUCTURE DEVELOPMENT

3.1 International experience of research infrastructure recovery

Rebuilding Ukraine's RI is an extremely challenging task as this means not just rebuilding after individual emergencies but a systemic better-than-before renewal after a long period of crisis and massive destruction.

Based on the case studies below, it becomes clear that Ukraine has to find its own way to recover RI on a large scale, as there is no recent experience to draw on. However, it is important to consider certain elements of international experience.

Case study of the University of California[73]. *An explosion in a University of California laboratory and the destruction of equipment demonstrates the importance of research requirements compliance and the need for preventative tools. A fire in a chemistry laboratory caused by the ignition of benzene vapour during distillation resulted in the injury of a doctoral student and caused approximately USD 3.5 million in damages, including repair/rebuilding costs[74]. The case is interesting for the assessment of scientific losses from the accident and the importance of having data backups (10 doctoral students working in the lab lost years of research due to the lack of research data copies). It would be useful to note the approach to renovating research buildings in accordance with modern building codes, including the installation of fire sprinkler systems.*

[73] Gottlief, J., Beatriz Cholo, A. (2001). Explosion in UC Irvine Science Lab Caused Less Damage Than Feared. URL: *https://www.latimes.com/archives/la-xpm-2001-jul-25-me-26368-story.html*.
[74] University of California, Irvine Independent Accident Investigation. Final Report. January 24, 2002. *https://www.prismrisk.gov/services/risk-control/toolbox/information-sheets/accident-investigation/uc-irvine-accident-investigation/*.

Tohoku University case study[75]. *The 2011 earthquake in Japan caused severe damages to many universities, resulting in suspension of classes in a number of regions, making hundreds of international students and scholars to leave the country, and leading to cancellation of international events. Among the universities most affected by the earthquake was Tohoku University, one of the world's top engineering schools with five campuses in the coastal city of Sendai.*

As a result of the earthquake, some buildings at the Aobayama Engineering and Science Campus were damaged and had to be demolished. In total, the earthquake resulted in 28 Tohoku University buildings (5% of the total) being declared "unsafe" and restricting the use of another 48 buildings (8% of the institute's buildings). After temporary reinforcement work was carried out on many buildings, inspections revealed that three buildings affected by the earthquake (the Civil Engineering and Architecture Campus; Electrical Engineering and Applied Physics; and Materials Science and Engineering) would have to be demolished or rebuilt.

Official estimates of the cost of the damage are difficult to come by, but Masahiro Yamashita, a chemistry professor at Tohoku, says the total damage is estimated at USD 1 billion, and additional funds are being allocated for recovery. Yoshihito Watanabe, vice president of Nagoya University and professor of bioinorganic chemistry, notes that a number of other universities have also been affected by the disaster, including Fukushima University and Tsukuba University. According to Watanabe, Japan's Ministry of Education has spent USD 1.6 billion on universities, including reconstruction of research facilities and scholarships. An additional USD 1.2 billion was subsequently allocated.

After the earthquake, a number of measures were taken in cooperation with local authorities to strengthen buildings. Among other things, Tohoku University combined reconstruction with the construction of a subway, which will make the campus earthquake-proof. Tohoku plans to build a smart earthquake-resistant building for scientists' work in the natural sciences.

These cases demonstrate a reactive approach to the restoration of the physical component of RI after a short-term crisis with the introduction of more effective post-facto preservation strategies. In our opinion, in the context of a full-scale and long-term war, the restoration of RI should be part of strategies for the renewal of scientific and innovation activities specifically and the country in general, implemented as a systemic policy during the crisis.

The historical example of one of the Middle Eastern countries, the Islamic Republic of Iran, illustrates recovery during crisis, as described from personal experience of one of the experts working with scientists in this country in the period up to 2014, and also based on an analysis of data from open sources. The country is authoritarian and poses a threat to the region and the world, including by collaborating with Russia and other terrorists, so its strategic approach to science and technological development is important to analyse. This is what makes its technological superiority possible and increases the threat to Ukraine and other democracies. In the context of war, long-term and large-scale losses of human capital and RI, Ukraine has to make special efforts to strengthen its technological capacity to win the war and recover as a strong and successful knowledge-based democracy capable of defending itself in the future.

Case studies of science recovery in the Islamic Republic of Iran

Case 1: Building educational and research infrastructure during the war. During the war

[75] Stafford, N. (2012, March 21). Quake-hit Japanese universities move on. URL: *https://www.chemistryworld.com/news/quake-hit-japanese-universities-move-on/3003355.article*.

which lasted for almost 10 years, a university was founded to train master's and doctoral programme students to meet the need for highly qualified teachers and researchers for universities and research centres in the country.

Case 2: Selection of national priority research areas. The country's top leaders were aware of the country's technological stagnation and the need for rapid technological upgrading. This resulted in the selection of a key applied research area which was supported by the necessary funding and contributed to the strengthening of basic research in the natural sciences, leading to a significant increase in the number of scientific publications in physical and chemical sciences and engineering.

Case 3: "20-year plan for knowledge-based economy". In 2009, a 20-year "comprehensive plan for the development of a knowledge-based economy" was adopted, focusing on science in higher education and research partnerships between industry and universities[76]. It accommodates the strategic and innovation priorities of various science and technology sectors[77]. In addition to increasing the number of scientists to 3,000 per million people, the plan sets a goal of increasing investment in research to 4% of GDP and 7% in education, 50% of which will come from the private sector. As of 2019—the midpoint of the plan—the number of scientists had grown to 1,700 per million people (compared to approximately 700 in 2012), and the allocation of research funds had increased to 0.9% of GDP (from 0.3% in 2012) and 3.6% of education funds (which has remained almost constant)[78].

Case 4: Investing in talent. In 2005, the National Foundation of Elites was established to identify, attract, and support talented society members[79]. The Foundation members are individuals who demonstrate exceptionally high intellectual potential, academic ability, creativity, and artistic talent, specifically contributing to the advancement of world science, and highly cited scientists and researchers. The Foundation provides support—financial/tangible and intangible, low-interest or free loans, provision of the necessary laboratory facilities, engagement of members in highly demanded/top priority national projects (in the case of young male members, instead of compulsory military service they have a change to implement their own research project in a scientific and technological environment), assistance in commercialising research results, and so on.

It is worth noting that military needs are often the key motivation for authoritarian countries to invest heavily in science. They often take a strategic approach to RI development. The development of the military-industrial complex in such countries is a key issue and requires constant (at times forced) involvement of researchers in the development of new types of weapons and means of warfare, as such countries are constantly preparing for war. Democratic countries need to pay attention to this in the context of their development of non-military sectors and the ratio of allocations between defence and civilian research. It also means that democracies at risk should prioritise the development of science, innovation and technology sectors in order to effectively defend and develop.

In Ukraine, even before the full-scale invasion, the number of employees involved in scientific research and development decreased from 182,484 to 78,860 scientists between 2010 and 2020 (of which the number of researchers was 133,744 in 2010 and 51,427 in 2020; the number of doctors of sciences decreased from 11,974 to 7,060, and the number of candidates of sciences decreased from 46,685

[76] 20 year national vision. Iran Data Portal. URL: *https://irandataportal.syr.edu/20-year-national-vision*.
[77] Sawahel W. Iran: 20-year plan for knowledge-based economy. University World News. Publ. Aug. 30, 2009. URL: *https://www.universityworldnews.com/post.php?story=20090827175231370*.
[78] Global Innovation Index 2022. URL: *https://www.wipo.int/edocs/pubdocs/en/wipo_pub_2000_2022/ir.pdf*.
[79] National Foundation of Elites. URL: *https://en.bmn.ir/*.

to 17,949)[80,81]. At the same time, the number of postgraduate students in 2022 almost reached the 2010 figure — 33,826 and 34,653 applicants, respectively — after a significant decline in the mid-2010s (22–25 thousand people)[82], which creates preconditions for the restoration of scientific personnel in HEIs and research institutions in the following years. The research intensity of Ukrainian GDP (R&D expenditures from all sources as a percentage of GDP) in 2019 was 0.43%, including 0.17% from the state budget[83], which is ten times less than the figure specified in paragraph 2 of Article 48 of the Law of Ukraine *On Scientific and Technical Activity*[84].

For Ukraine as a democratic state that serves as a model in the struggle for its values, the following may be important in the experience reviewed:
- research infrastructure should be developed even in times of war;
- technological capabilities should be developed, including through dedicated support of relevant priority areas of research with simultaneous support of related abstract science areas;
- a national plan for the development of a knowledge-based economy is needed;
- effort should be made to increase the number of scientists and investment in research and education, including creating conditions for an increase in the share of revenues from the private sector;
- implementation of the principle “a scientist is the elite of the state”, which contributes to raising the social status of scientists, special support for leading and young scientists, and creation of the necessary conditions for their activities;
- development of mechanisms for closer interaction between university/academic science and applied research in the military sphere.

3.2 Research infrastructure in regional development and smart specialisation

Smart specialisation is one of the tools that link human capital, scientific potential, and RI in the development of systemic local development strategies.

Establishment of a new RI is a long-term process that lasts at least 10 and sometimes 15 years: it consists of making coordinated political decisions, including on cross-sectoral and interstate coordination of preparatory actions, drawing up a plan of such actions, and strict adherence to the timeframe for its implementation, conducting a technical and economic assessment and drawing up a feasibility study on its basis, developing (if necessary) technical documentation and designing facilities, constructing them, filling them with the necessary equipment, including

[80] Number of employees engaged in scientific research and development by personnel category (2010–2020). State Statistics Service of Ukraine. URL: *https://ukrstat.gov.ua/operativ/operativ2021/ni/rik/kpzvndrkp_10_20_ue.xlsx*.
[81] Number of employees with an academic degree engaged in scientific research and development (2010–2020). State Statistics Service of Ukraine. URL: *https://ukrstat.gov.ua/operativ/operativ2021/ni/rik/kpzvndrns_10_20_ue.xlsx*.
[82] Number of postgraduates. State Statistics Service of Ukraine. URL: *https://ukrstat.gov.ua/operativ/operativ2022/osv/aspirantura/Kilxk_asp.xlsx*.
[83] Scientific and technical activities in Ukraine in 2019. Scientific analysis and report / T.V. Pysarenko, T.K. Kuranda et al. K.: UkrINTEI, 2020. 109 pages. URL: *https://mon.gov.ua/storage/app/media/nauka/informatsiyno-analitychni/2020/08/13/nadnaukaza2019-stisnuto.pdf*.
[84] Law of Ukraine On Scientific and Technical Activity. The Official Bulletin of the Verkhovna Rada, 2016, #3, p.25. URL: *https://zakon.rada.gov.ua/laws/show/848-19#Text*.

technical support (high-voltage power grids, communications of various kinds, and more). Obviously, such a significant volume of diverse activities requires local resources and thus contributes to the economic development of the region, including the provision of a large number of services, both long-term and short-term. In turn, businesses are engaged in the establishment of new RI, which also has a positive impact on regional development and is directly or indirectly related to smart specialisation.

Research and innovation strategies for smart specialisation (RIS3) are based on the analysis of the regional economy, society, and innovation structure to assess both existing assets and prospects for future development. The analysis covers four main dimensions: regional assets, including RI, human capital quality and specialisations, the region's connections and role in the European and global economy, and the dynamics of the business environment. The RIS3 aims to support policies and investments in key national/regional (EU) priorities for knowledge-based development with high potential to address societal challenges[85].

The potential of RI within smart specialisation strategies can be developed within macro-regional strategies. The EU's Macro-Regional Strategy (MRS) is a policy framework that allows countries located in the same geographical region to jointly seek solutions to problems or better use shared capacities when addressing issues across national borders (such as environmental pollution, shipping, global business competition, and more). According to a factsheet published by the European Commission, in this way they benefit from enhanced cooperation to make their policies more effective than in cases if problems were solved individually[86].

For example, the role of RI was taken into account in the Interreg Danube Transnational Programme projects[87].

The project "Facilitating macro-regional scope and link up to socio-economic actors of Research Infrastructure in the Danube Region" (RESINFRA@DR, 2017–2019) aimed to improve basic conditions for research infrastructure and innovation in the Danube Region and to focus them on relevant regional, national, and macro-regional policy areas, as well as at the EU level, with the involvement of RI end-users. The project envisaged tailored trainings for the identified target groups, thus contributing to capacity building covering the entire macro-region and beyond the involved partnerships; the establishment of a pool of qualified reviewers for the assessment of RIs; pilot activities focused on peer review of nine existing RIs and preliminary assessment of three planned RIs in the macro-region.

Another project, "Embeddedness of high-quality research infrastructures in the Danube Region" (RI2INTEGRATE, 2017–2019), was aimed at harnessing the economic development potential and better integration of the work of EU investment projects in the field of research infrastructure (RI) by developing and implementing innovative tools for policy studies on macro-regional embeddedness in the Danube Region. Three tools were developed and tested to enhance macro-regional RI embeddedness (public procurement guidelines for innovation; a guide to assist the RI-related business ecosystem; and a roadmap for community engagement). To ensure policy sustainability, seven national and one joint action plans have been developed, and a transnational RI2integrate committee will be funded to cover all Danube Basin countries. The novelty of the RI2INTEGRATE project included two main aspects. Its methodology involved a combination

[85] Clar, G. (2018). Guiding investments in place-based development. Priority setting in regional innovation strategies. Seville: European Commission, JRC112689. URL: *https://joint-research-centre.ec.europa.eu/publications/guiding-investments-place-based-development-priority-setting-regional-innovation-strategies_en*.
[86] The ABC of Macro-Regional Strategies. An EUSDR Implementation Toolkit. URL: *https://danube-region.eu/wp-content/uploads/2022/12/EUSDR-Tool-kit_final_dec-2022.pdf*.
[87] Approved projects. Danube Transnational Programme. URL: *https://www.interreg-danube.eu/approved-projects*.

of a smart specialisation approach (from the experts' side) and a four-helix model (from the stakeholders' point of view). In addition, the project focused on synergies between different EU funding instruments and territories.

The RESINFRA@DR and RI2INTEGRATE projects are important for Ukraine and can serve as a source of useful conclusions and recommendations for the development of RI and the innovation environment, including through participation in macro-regional programmes and the formation of effective RI development policies. It is important for Ukraine to join or initiate similar projects in the future.

An example of a network approach is the Vanguard Initiative[88], an alliance of 38 of Europe's most advanced industrial regions focused on fostering industrial innovation and building European value chains вартос-ті based on complementarities in regional smart specialisation strategies. By connecting innovation ecosystems and sharing knowledge and tools between participating regions, the Vanguard Initiative promotes interregional cooperation, stimulates cross-regional investment in innovation, strengthens open innovation, and accelerates the introduction of new products and innovations to the European market.

Through its pilot projects, the Vanguard Initiative has proven to be a successful network that sets an example in interregional cooperation based on smart specialisation. The Vanguard Initiative is linked to the work of the EC's Smart Specialisation Thematic Platforms (S3Ps), advocating for stronger commitments to ensure effective synergies between EU funding instruments and regional and national funding.

For example, Phase 2 of the ProNano project brings together regional infrastructures within the Vanguard network with a focus on nanotechnology industry development, considering the needs of EU industry and SMEs. Located close to the ESS and MAX IV research infrastructures in Lund, ProNano aims to complement other European nanotechnology infrastructures[89].

The case of the Tigray region (northern Ethiopia) illustrates the role of RI in addressing regional development issues, as well as a comprehensive approach to post-crisis agricultural sector recovery involving RI.

The Tigray region is an example of building a successful model for land restoration in Africa that has received numerous awards and recognition. Since 1991, impressive rehabilitation has taken place, including re-vegetation, reduction of rainwater runoff, and improvement of soil quality. The availability of fodder, food, and fuel has greatly increased, and forest cover has grown from 3% to 17%[90].

In 1999, the Tigray Agricultural Research Institute (TARI) was established to support regional agricultural development policy based on sustainable development in Tigray to develop and promote innovative agricultural technologies and practices.

The two-year war and siege in the Tigray region (2020–2022) undid many efforts to restore the Tigray ecosystem. The conflict has had a devastating impact on both people and the environment.

RIs in Tigray, such as genetic resources, laboratories, agricultural machinery, and nursery

[88] Approved projects. Danube Transnational Programme. URL: *https://www.interreg-danube.eu/approved-projects*.
[89] Progress update from the Nano Pilot. Vanguard Initiative. 09.07.2021. URL: *https://www.s3vanguardinitiative.eu/news/progress-update-nano-pilot-0*.
[90] Gebrekirstos, A. (2023, April 14). The war on Tigray wiped out decades of environmental progress: how to start again. The Conversation. URL: *https://news.yahoo.com/war-tigray-wiped-decades-environmental-122732267.html?guccounter=1&guce_referrer=aHR0cHM6Ly93d3cuZ29vZ2xlLmNvbS8&guce_referrer_sig=AQAAAEENDiyq3RzIcGeTBNPVbp6c6-cQzom_W_UA7GJeZWHsu-FBqKhKKjOTlAqEGX03M_1YRY5U7y0_COFhTa5aPDxvKvQI5SKQefp8hRT1F3OmPheSfLVFRGebfnrE2-fNak2J3AqhxqgqiiQXXegXJ0_SvJgoridSobmtabh8Yfm0*.

infrastructure, were destroyed or looted. TARI, which has made perhaps the greatest contribution to the rehabilitation of Tigray lands, was also destroyed. A damage assessment estimated the value of TARI's losses at approximately USD 75.5 million. Genetic resource repositories, laboratories and field equipment, agricultural machinery, IT equipment, and data centres were destroyed. All research programmes, projects, and experiments, including 39 doctoral projects, were lost[91]*.*

The resumption of TARI's activities is planned to be carried out under a comprehensive strategy with the support of national and international sources[92]*. A project by Agence Française de Développement (AFD) and the EU aims to help repair the damage, restore Ethiopia's agriculture, and put the country back on track for food security, with AFD providing EUR 18 million in grants and the EU delegating an additional EUR 14 million to AFD until 2026. All this is part of the FARM project "Food Security and Agricultural Recovery Measures in Conflict-Affected Regions of Ethiopia".*

To strengthen the agricultural sector in the long term, in addition to immediate interventions in conflict-affected areas, the project also aims to increase Ethiopia's food self-sufficiency. The country is dependent on imported resources, such as fertilisers and agrochemicals, which makes it vulnerable to market fluctuations and external shocks. As part of the recovery, it is planned to establish an agricultural marketing cluster aimed at transforming agricultural business models. To this end, the project plans to provide integrated support to the seed sector through strengthening agricultural R&D to develop agricultural technologies adapted to different agro-ecological zones and climatic conditions that few farmers currently have access to. It is planned to involve all stakeholders and development partners in the chain, including research centres, agricultural offices, seed production companies, NGOs, and other public and private sector representatives. This will include rehabilitation of conflict-affected institutions, as well as seed research, production, multiplication, and soil analysis. Along with this, there are plans to train farmers in agro-ecological practices, enabling them to take advantage of innovative agricultural technologies, natural resource management techniques, promote the use of organic fertilisers, and sustainably improve agricultural production and productivity.

The next case study of RI as a factor of smart specialisation concerns the recovery of the pearl farming industry in Fiji after Cyclone Winston (2016)[93].

The case demonstrates how the aquaculture RI, crucial for the development of the industry, can draw the attention of international donors for rehabilitation. The monitoring of the destroyed RI was carried out in cooperation with scientists, the government, and businesses. The assessment of key RI needs was used as a basis for equitable distribution of infrastructure recovery assistance. The Secretariat of the Pacific Community (SPC) was consulted during the initial planning of the studies, which enabled coordination of subsequent donor activities (such as the EU-funded Increasing Agricultural Commodity Trade (IACT) project). The RI was restored as part of a project that included the following stages:
1. Meetings with staff to finalise specific needs in equipment and materials.
2. Liaising with businesses and communities to identify locations for equipment to be deployed and deployment timeframe.
3. Sourcing appropriate equipment from suppliers.

[91] World peace foundation (March 1, 2022). Tigray: The Destruction of Invaluable Agricultural Research. URL: *https://sites.tufts.edu/reinventingpeace/2022/03/01/tigray-the-destruction-of-invaluable-agricultural-research/*.
[92] Ethiopia: reviving agriculture in areas afflicted by conflict. AFD. 03 April 2023. URL: *https://www.afd.fr/en/actualites/ethiopia-reviving-agriculture-areas-afflicted-conflict*.
[93] Australian Centre for International Agricultural Research (2017). Pearl industry research infrastructure recovery post cyclone Winston. Final report. URL: *https://www.aciar.gov.au/sites/default/files/project-page-docs/final_report_fis-2016-005.pdf*.

4. Installation of equipment at pearl farms.
5. Controlling the recruitment and training of personnel to operate the installed equipment.

The main outcomes of the RI restoration project:

- *rapid response to the loss of RI (about a year passed between the first consultations with farmers after Cyclone Winston (March 2016) and the first harvest of oysters) minimised the period of loss of results and business losses;*
- *Fiji's pearl industry has become more resilient through increasing capacity, spreading operations over a wider geographical area, standardising culture/methods and expanding operations to new communities that cooperate with pearl farmers;*
- *greater resilience was achieved through the deployment of new equipment that is more robust and resistant to damage from future cyclones and storms;*
- *introduction of the new RI has influenced the expansion of the industry, leading to increased pearl production and revenue from sales on the national market and from exports;*
- *ties have been established with international equipment suppliers, which contributes to increased capacity to support future expansion of the industry;*
- *communities involved in the re-establishment of sectoral RI worked directly with project staff to plan, deploy, and maintain the new research infrastructure, resulting in significant capacity building;*
- *extensive media coverage of the project raised the prestige of RI and the industry in Fiji.*

Experience of implementing smart specialisation in Ukraine and RI

Ukraine faces the need to consider the issue of RI in the context of restoring the affected areas. The use of the smart specialisation approach in the process of restoring the territories helps make this process more systematic, scientifically justified, and efficient.

In 2016, discussions on the possibilities of introducing smart specialisation in Ukraine were initiated and an active dialogue with the EU was launched along with an information campaign by the Joint Research Centre (JRC) to raise awareness of new opportunities arising from Ukraine's scientific and technical cooperation with the EU in the context of introducing smart specialisation. In 2017, the implementation of smart specialisation was launched in three pilot regions of Ukraine: Kharkiv, Zaporizhzhia and Odesa regions. In 2018, the Cabinet of Ministers of Ukraine adopted a Resolution on Amendments to the Procedure for Developing the State Strategy for Regional Development of Ukraine, a Resolution on the Procedure for Developing Regional Development Strategies and Action Plans for their Implementation, as well as Monitoring and Evaluating the Implementation of the said Regional Strategies and Action Plans, and an Order of the Ministry of Regional Development on the Methodology for Developing, Monitoring, and Evaluating the Implementation of Regional Development Strategies and Action Plans for their Implementation[94]. On 4 August 2023, the CMU adopted resolutions *On Several Matters of Drafting Regional Development Strategies and Action Plans for Their Implementation and Monitoring the Implementation of the Same*[95] and "*On Several Matters of Developing the State Strategy for Regional Development of Ukraine and an Action Plan for its Implementation and Monitoring the Implementation of the Said Strategy and Action Plan*[96], which update the mechanisms for drafting regional development strategies. These documents have a framework

[94] Order No.70 of the Ministry of Regional Development, Construction, Housing and Communal Services of Ukraine On Approval of the Methodology for Development, Monitoring, and Evaluation of the Effectiveness of Implementation of Regional Development Strategies and Action Plans for their Implementation of 31 March 2016. URL: *https://zakon.rada.gov.ua/laws/show/z0632-16#Text*.
[95] Resolution No.816 of the Cabinet of Ministers of Ukraine On Several Matters of Drafting Regional Development Strategies and Action Plans for Their Implementation and Monitoring of Implementation of the Same of 4 August 2023. URL: *https://zakon.rada.gov.ua/laws/show/816-2023-%D0%BF#Text*.
[96] Resolution No.817 of the Cabinet of Ministers of Ukraine On Several Matters of Developing the State Strategy for Regional Development of Ukraine and an Action Plan for its Implementation and Monitoring the Implementation of the Said Strategy and Action Plan of 4 August 2023. URL: *https://zakon.rada.gov.ua/laws/show/817-2023-%D0%BF#Text*.

nature, so when determining the strategic priorities of specific regions, the RI as a resource should be considered at the expert level.

During the focus group with experts—representatives of CSUSEs of HEIs and the National Academy of Sciences of Ukraine—it was noted that the existing RI of Ukraine is not currently included in regional development programmes (projects). The participants also stressed that local society is unaware of the availability and capabilities of RI. The experience of the authors of this paper and communication with experts shows that RI has not been considered within the development of smart specialisation strategies. At the same time, the idea of linking RI to smart specialisation strategies is supported at the expert level.

Yevhen Anhel, Senior Research Assistant at the Institute for Economic Research and Policy Consulting, noted in an interview the expediency of creating regional RI centres (for instance, Lviv, Kyiv, Odesa, Dnipro, and Kharkiv). Some regions of Ukraine have powerful institutions working with single-purpose topics. Accordingly, a RI centre could be established and developed at their facilities, purely in their field. Speaking of large cities, several sectoral centres may be created. For example, an agricultural RI cluster could be established in Cherkasy region, given the strong educational and scientific agricultural cluster, whose members are quite successful in developing plant varieties, etc.

Iryna Pidorycheva, Head of the Sector at the Institute of Industrial Economics of NASU, noted the need to consider the RI aspect within smart specialisation strategies to create qualitatively new advantages for regions.

During the expert interviews, Andrii Vasyliev, director of Briolight-Ukraine LLC (a knowledge-intensive startup), associate professor at the Petro Mohyla Black Sea National University, presented the case of establishing a laboratory through the unification of regional centres, which the regional authorities allowed and even helped to create. At the same time, the instability of financial support was emphasised. In Ukraine, an example of the implementation of RI-based regional development projects is the establishment of the agritourism-focused hemp breeding cluster "Slobozhanske Konopliarstvo" funded by the State, regional budgets, and other sources. The Sumy region, with its strong scientific potential and good conditions for growing hemp, has every opportunity to be competitive at the global level.

The project envisages the creation of a national Ukrainian training and practical centre for the development of hemp breeding with the active participation of the Institute of Bast Crops of the National Academy of Agrarian Sciences of Ukraine, the Institute of Agriculture of the North-East of the National Academy of Agrarian Sciences of Ukraine, establishment of the Centre for Ukrainian Hemp Breeding in Sumy, a mobile exposition on hemp breeding, etc. In addition to the two scientific institutes of the National Academy of Agrarian Sciences of Ukraine, the Sumy National Agrarian University which has a shared use centre, is involved in the implementation of projects within the Slobozhanske Konopliarstvo cluster. The centre has been modernised with scientific equipment and adapted for hemp research. Three scientific laboratories have already been re-equipped, and the fourth will be specialising entirely in hemp breeding. In 2021, the Slobozhanske Konopliarstvo Cluster was selected by Sumy Region as the best project with cluster potential to increase competitiveness as part of the GIZ Ukraine project "Application and Implementation of the EU-Ukraine Association Agreement in the field of Trade / EU4Business: SME Competitiveness and Internationalisation".

This case study may serve as an important example for other regions of Ukraine that also have scientific potential, including RI, and opportunities for the development of certain industries. It demonstrates how the joint work of scientific institutions, businesses, and the government can lead to the establishment of a successful cluster and contribute to the development of the region and its RI. Moreover, this case study shows the importance of

engaging external funding sources, such as international organisations projects, to implement such initiatives.

Based on the analysis of the inclusion of RI development in development projects within regional and macro-regional EU strategies, we believe that the development of RI in Ukraine should be considered as the basis for building/updating smart specialisation strategies, particularly in the development of regions that will be reintegrated and/or were significantly affected by military operations, as a result of which the restoration of old industrial models is impractical.

3.3 Ukrainian experience and promising ideas for the development of research infrastructure

During the focus groups and expert interviews, the authors of this paper received several ideas and suggestions for the development of RI from the respondents.

The respondents noted the need for cooperation with businesses in the development of RI. Andrii Vasyliev, director of Briolight-Ukraine LLC (a knowledge-intensive startup), associate professor at Petro Mohyla Black Sea National University, notes the relevance of opening joint laboratories (business–Ukrainian university–foreign university). Under this model, a business buys equipment, a Ukrainian university provides premises, and a foreign university attracts specialists. This results in a mini-consortium able to raise additional grant funding. Regarding the idea of building a national RI model, the expert suggests that an open access model based on a nationwide franchise is quite promising. This would encourage businesses to invest in RI in certain HEIs/SIs, and allow users to access such centres at the expense of grant funds. The expert proposes differentiating funding for creating a public product (80% public funds, 20% private funds) and applied research (80% business funds, 20% state funds). Additionally, the expert emphasises the importance of professional administration of the model's operation.

Due to existing problems, particularly in the interaction between RI and businesses, new legal forms of RI have been proposed. Specifically, Oleksiy Boyarsky, a representative of CERN (Switzerland), emphasized the importance of creating a transparent and clear type of a RI organization as a scientific non-profit organization with transparent access rules, a supervisory board, attractive tax status, and government support.

Olena Orliuk, Director of UANIPIO, noted the expediency of establishing thematic RI centres as a basic way to preserve scientific potential, justifying this by the fact that leading universities have a more widely known status, so businesses and foreign donors will quickly enter partnerships with them. The expert notes that the development and use of RI is a matter of building good partnerships between universities and research institutions with groups interested in laboratory equipment. Cooperation between science and business, which has its own RI and more opportunities to upgrade it, may increase the efficiency of scientific research.

During the expert interviews, the idea of establishing specialised RI centres with the involvement of businesses and prioritising them was suggested. This task has the potential to significantly impact the development of science, technology, and the economy in a positive way. This idea proposes cooperation between HEIs/SIs to facilitate the exchange of knowledge and technologies, increase investment in research and development, promote the development of high-tech industries and regional innovation ecosystems, and provide access to modern technologies.

CSUSEs' functioning models

One idea for the development of RI in Ukraine is to transform the leading sectoral CSUSEs under the principles of operation of research and education centres. These centres would operate as knowledge hubs based on EU RI practices. When developing both national and local strategies for establishing RI centres and implementing RI monitoring, it is recommended to consider the use of research infrastructure management systems (RIMS). These systems should be in the format of a cloud-based online research infrastructure and a reservation system that ensures identification of researchers and their access to research facilities, equipment, and infrastructure that will provide data for a more effective management of RI use.

These ideas were formalised in the concept of a sharing CSUSE with the involvement of experts. The concept envisages the creation of an ideal model of RI operation with a self-management system where the organisation of RI usage will be digitised to optimise the use of premises, devices, equipment, and human resources for effective research activities.

The CSUSE model developed jointly with the respondents as a sharing centre can be based on the use of modern approaches to the research facilities design (availability of a reception point, automatic digital access to laboratories and equipment with an electronic key, etc.) The effectiveness of such RI is assessed on the basis of a clear digital footprint of the use of premises, devices, and equipment. There are two indicators — workload and rate of return of the infrastructure (paid access).

In addition to the laboratories, there may also be meeting rooms for negotiations with investors, patent, science, and insurance brokers, scientists, customer representatives and business startups. These rooms should provide a quiet and private environment, separate from the main research activities. They have to be designed flexibly enough to be subdivided into smaller rooms or combined into larger ones for joint meetings, seminars, conferences, and knowledge sharing.

The organisational model of CSUSE as an open space sharing centre can ensure efficient use of space, convenient access to resources and a comfortable environment for researchers, staff, and visitors.

To set up a CSUSE as an open space sharing centre, one needs to consider several key elements, as outlined in the research management literature. There has to be a relevant topic, an effective group of scientists, including technical staff, and a leader[97]. It is also important to ensure effective communication between researchers. In open laboratories, researchers share not only the space itself, but also equipment, workspace, and technical staff. The open lab format facilitates communication between scientists and makes the lab more easily adaptable to future needs[98].

Space design in scientific research centres can have a major impact on productivity,

[97] Owen H. A Brief User's Guide to Open Space Technology. URL: https://openspaceworld.org/wp2/hho/papers/brief-users-guide-open-space-technology/.
[98] Daniel Watch D., Tolat D. Research Laboratory. 2017. URL: https://www.wbdg.org/building-types/research-facilities/research-laboratory.

recruitment, morale, and retention of researchers and staff. Laboratories are where researchers spend most of their time, and the environment reflects the perceived value of the work and the people involved[99]. Besides, effective research centres must be appealing, inspiring, informative, easy to navigate, adaptable, efficient, technology-friendly, comfortable, quiet, and durable.

This means that space design at research centres is crucial to research efficiency and satisfaction of researchers. Therefore, it is important to pay attention to space design when planning a CSUSE to ensure optimal conditions for research and contribute to the success of the centre. Organising a CSUSE according to the proposed logic (Table 3) offers several advantages.

Furthermore, various studies demonstrate that open space offices have material benefits in terms of employee satisfaction and improved communication[100]. One of the greatest advantages of using open office space is that it reduces rental costs and increases employee productivity. Open office space can also facilitate better communication between employees and improve teamwork.

These centres are managed through self-management and digital systems. The digital format enables daily coordination of supplies, commissioning, repairs, service, and support. Decisions on improving the centre's processes, ordering additional equipment, and its development are made by delegated representatives of all participants in the centre based on quarterly meetings to increase productivity and efficiency

Table 3. Specifics of CSUSE operation in the system of MESU, NASU, and sharing CSUSEs

##	Name of activity area	CSUSE in MESU system	CSUSE in NASU system	Sharing CSUSEs
1.	*Equipment*	Funds were only allocated for equipment purchase in 2018–2019	Centralised procurement programme for expensive imported equipment and establishment of CSUSE for 2004–2010, from 2011 onward — repairs and modernization of equipment with much smaller financial opportunities	Constant updating in accordance with user requests covered by top priority state support at the stage of the centre's formation
2.	*Financial support of maintenance*	Not available	Insufficient	Provided by stakeholders based on a self-sustainability model (becomes possible through centre's promotion and engagement of users)
3.	*Online operations for reservations, reporting, and access*	Fragmented	Not Available	Working 24/7
4.	*Operating staff*	Scientific and technical (at the expense of a HEI's dedicated fund)	Scientific and technical	Scientific and technical

[99] Paul, G., Watch, D. (2017). Interior Design For Research Facilities. Excerpted from: Building Type Basics for Research Laboratories, Second Edition. URL: https://www.wbdg.org/resources/interior-design-research-facilities.
[100] 5 reasons to have an open-space office. 2021. URL: https://www.thimble.com/blog/open-office-benefits.

##	Name of activity area	CSUSE in MESU system	CSUSE in NASU system	Sharing CSUSEs
5.	*Business model for sustainable development*	Mostly private and individual, paid access upon decision of HEI administration	Access for NASU institutions is free of charge; third-party users pay for access	Open for everyone, user-friendly for paid access for everyone and free access for certain scientists
6.	*Reporting*	Upon inquiry on paper, manual	Annual reporting pursuant to the regulatory documents of NASU	Automatically generated using data from the access platform
7.	*Tenders to receive equipment*	Not available, private distribution	Not available, upon inquiry	Upon inquiry, public-private partnership (including brand franchise), grants

Research centre leaders, who are usually well-known researchers in their field, are not involved in administrative matters and play an important role in coordinating researchers[101]. A centre leader defines the goals and vision of the development area, identifies resources, monitors processes (including finance, communications, IT), and develops a plan for achieving the goals.

Such centres receive funding from several sources:

- individual payments by researchers for access to the centre based on a personal card;
- payments for usage by a certain group of researchers;
- institutional funding;
- payment for services by businesses;
- donor funds for the development of research infrastructure;
- provision of equipment by manufacturers to promote their brand among researchers;
- state subventions and grants;
- startups of funding individual investors and funds.

Such centres should be funded in two stages and from different sources:

1. key capital funding — from the state budget or donor funds;
2. operating costs and support for the sustainable development of the centres — at the expense of their income-generating activities, the products of which should be created and managed by the professional administration of such centres. The local budget should also support the operational activities of such a centre if they see benefit from its activities. The CSUSE should be involved in the implementation of smart specialisation in the region.

The restoration of Ukrainian RIs cannot be based on the "a little bit for everyone" principle. It would be advisable to identify local territorial and human capacities to create an economic cluster based on the principles of smart specialisation, scientific and educational institutions and organisations (equipment, devices, working conditions, personnel), and then build no more than one sharing research centre per three administrative regions on its basis. It will function as a research infrastructure that, along with research and service provision, will contribute to the creation of innovative products necessary for the post-war effective economic recovery of Ukraine.

The development of RI by HEIs/SIs using their own resources is also an important

[101] What does a research leader do? Zippia. https://www.zippia.com/research-leader-jobs/what-does-a-research-leader-do/?src=chatbot_popout_displayed.

task for increasing research potential and competitiveness. There is a positive experience presented at the focus group regarding the development of a CSUSE at the Sumy State University based on the following components which ensure the development of the logistics of the Centres:

1. participation in all competitions that enable infrastructure development — NATO projects, NRFU, Horizon Europe, projects of basic funding for scientific activities of the Ministry of Education and Science of Ukraine, among others;
2. raising funds through the execution of business contracts;
3. attracting sponsorship;
4. advertising the opportunities of CSUSEs to stakeholders.

Sectoral model of product-oriented RI development

An example of the RI development concept with consideration of industry trends and product orientation, specifically for defence purposes, is found in the report of Vasyl Kladko (the scientist was killed by the Russian occupiers in Irpin on 13 March 2022), Deputy Director for Research at the Lashkaryov Institute of Semiconductor Physics of the National Academy of Sciences of Ukraine[102]. According to the scientist, the following measures should be taken to develop the microelectronics industry in Ukraine, considering the potential of the available RI:

1. establish a research and education technology centre under the joint jurisdiction of NASU and MESU;
2. engage enterprises that still possess certain technological and human resources in its activities;
3. engage private businesses in the development and manufacture of specific products under investment agreements.

To improve the state of the microelectronics industry in Ukraine, scientists are proposed the following solutions to the problem: the establishment of a state key laboratory at the facilities of the Lashkaryov Institute of Semiconductor Physics of NASU, the principal activity of which should be the development, production (at the global technical level), comprehensive testing and supply of functional microelectronic components, including infrared components, for the needs of the national defence industry, as well as for export to other countries; the Lashkaryov Institute of Semiconductor Physics of the National Academy of Sciences of Ukraine should reconstruct the premises necessary to set up production lines, install appropriate equipment and measuring instruments; form a scientific and technological department of materials science as a branch of the key laboratory to organise real production and supply of epitaxial structures for the manufacture of electronic components based on silicon, indium antimonide, and trinitrides; set the approximate number of scientific and technical staff of the state key laboratory at 300–400 people and provide them with the necessary basic funding from the state budget; develop and approve the State Target Scientific and Technical Programme of the State Key Laboratory for 5–7 years with guaranteed budget funding aimed at ensuring national security and significant expansion of Ukraine's export potential in the high-tech field; create conditions for raising funds from sponsors, patrons, and patriots of Ukraine.

This case study is valuable for the rehabilitation of Ukrainian RI as it provides specific and detailed recommendations for the development of the microelectronics industry based on value chain design, starting with the existing RI. The implementation of these recommendations can help create not only a stable and competitive microelectronics

[102] Kladko, V. P. (2022). On the establishment of a state key laboratory for the development of microelectronics in Ukraine (transcript of report at the session of the NASU Presidium on 24 November 2021). Official Bulletin of the National Academy of Sciences of Ukraine, 1, 38-44.

industry in Ukraine but also contribute to national security and expand the country's export potential. This case study can serve as an important source of information for governmental and research institutions seeking to improve the state of RI in Ukraine.

V. P. Kladko's ideas were developed by his colleagues at the Lashkaryov Institute of Semiconductor Physics of NASU and supported by the relevant decision of the NASU Presidium[103].

[103] Resolution No.314 of the NASU Presidium of 6 September 2023 On Restoration of the Semiconductor Industry in Ukraine: Problems and Solutions.

SECTION 4. RECOMMENDATIONS

1. Improving the efficiency of state management of RI

The restoration and development of RI should be carried out under general strategies for the post-war renewal of innovative development, meet the priorities of science and technology development, as well as the needs of post-war recovery, and be carried out through the integration of innovative technologies and resources, international cooperation, and the establishment of effective research centres to address current challenges and promote the development of the economy and society.

One of the fundamental issues is the development of the State Target Programme for the Development of Research Infrastructures in Ukraine until 2026 (or for a new period) on the basis of the relevant Concept, considering the priorities for the development of science and technology updated in the context of post-war recovery, as well as best practices in the drafting of RI development roadmaps. At the same time, before starting this work, it is imperative to first carry out monitoring of the existing elements of RI, conduct an in-depth analysis of the use of their capabilities and potential, as well as model and discuss options for their operation with a range of stakeholders.

The effectiveness of the unified system of state management and coordination of the processes of RI operation should be ensured by an interagency coordination centre (the format is defined by MESU as the CEB in the field of science), which should disseminate the best RI operation practices, ensure the functioning of the relevant section “Research Infrastructure” of the specialised web portal nauka.gov.ua, which will contain information on available RIs, their capabilities and contact information to organise access, in particular through the use of RI management systems (RIMS).

For the purposes of RI development, it is recommended to hold competitions for NRFU funding in the field of development of material and technical base and research infrastructure. These competitions should involve foreign donors and have a clear definition of the objects of support, including the implementation of RIMS, digitisation of collections, and other relevant areas. The

government should promote the use of NRFU competitions as a transparent and effective tool for utilizing foreign aid. When conducting NRFU competitions to support mobility, the government should consider involving Ukrainian scientists in the world's leading RIs.

To gain a competitive advantage, it is necessary to improve the possibilities of forming public-private partnerships, such as investment agreements and franchises, to allow private capital to finance Ukrainian science following the example of the United States. Currently, the state budget of Ukraine does not have sufficient funds.

2. Ensuring the resilience of RI against risks and threats.

The requirement to ensure resilience and preservation of RIs during unforeseen circumstances that carry threats of loss or damage should be incorporated into the architectural and technical planning of their construction and placement. It is also advisable to consider the possibility of using technical solutions that enable urgent relocation of large equipment. The Model Provision on the Centre for Shared Use of Scientific Equipment (Resolution No.703 of the Cabinet of Ministers of Ukraine of 21 June 2022) and the Regulations on CSUSE should provide an algorithm for the preservation (evacuation) of scientific equipment of the Centres in the event of unforeseen circumstances and military actions. Additionally, a reserve fund must be maintained to ensure the practical implementation of the algorithms when necessary. Further details on this matter can be found in the paper Preservation.

3. International aspect of RI recovery and development

To enhance the participation of HEIs/SIs in the EU Framework Programmes for Research and Innovation which support RI development, to learn from the experiences of other countries, obtain necessary information directly from the European Commission and then communicate it to stakeholders in Ukraine, MESU should begin with officially nominating the NPCs of Horizon Europe, ensuring their support, and facilitating cooperation between the Horizon Europe Office in Ukraine, the NPC network, and Ukrainian representatives in the committees of this programme.

Support programmes for researchers should be put in place for the provision of laboratory facilities, including access in hybrid formats, to the research infrastructures of the European Research Area. It is advisable to establish an inter-institutional postgraduate (doctorate) programme to train researchers in specific fields of knowledge and specialties for all HEIs and research institutions. Additionally, the content of the Presidential University project should be adjusted in terms of the second and third level higher education organization, with an enhanced research component and access to advanced Ri in Ukraine and abroad.

4. Improvement of CSUSE operation

For a fast restoration of Ukraine's RI, it seems most appropriate to attract funding from international donors to reform CSUSEs and ensure their operation, which will contribute to the comprehensive development of the research system based on an approach that includes the following provisions:

- CSUSE network — clear and transparent rules and procedures for regulating the distribution of equipment between the Centres;
- guaranteed availability — providing access to equipment to a wide range of researchers, stimulating the use of CSUSE resources in the implementation of research and development projects;
- digitalisation of activities — electronic reporting, IoT technologies for automatic monitoring of equipment performance parameters and condition;
- staff — scientific and engineering personnel with competitive salaries, their training and retraining to ensure efficient use of equipment and user consultations;

- a CSUSE coordination centre and new management standards — a unified system for managing and monitoring activities, a digital management system, online equipment booking, customer and consumables records, monitoring the technical condition of equipment, and virtual access;
- inancial support — stable funding and regular investments in infrastructure upgrades, development, and maintenance.

To ensure the sustainable functioning of CSUSEs, the budget programmes of MESU and NASU should include general fund expenditures to support the operation of CSUSEs which will be used to maintain administrative staff for scientific and organisational work, engineering staff to provide maintenance of research-intensive equipment, advanced training of scientists in the use of modern scientific equipment, and purchase of consumables and components. At the same time, it should be not just the allocation of funds but comprehensive support programmes with specific target indicators (measurable, specific, and tied to the country's short- and medium-term development priorities).

5. Development of cooperation between HEIs/SIs/libraries and registers to accumulate primary research data for further sharing

The architecture of one of the future subsystems of the National Electronic Research and Information System (URIS) should have five structural levels:

- several registers of primary research data, patents, scientific property, research institutions, scientists, and more;
- a secondary research data base with specific IDs linked to common register codifications;
- research databases should be cloud-based, but the cloud should be formed in a hybrid format, where private clouds can be used locally to accumulate internal data and operational productivity of a particular research institution, its intellectual property, and public clouds where payment is only possible only for time and volume of use can be used in case of load surges during intensive joint projects and data exchange, if interaction buses are not provided (if interaction buses are provided, inter-sectoral and interdisciplinary ones can be formed);
- the platform infrastructure can be national or international, following the model of the European Science Cloud;
- e-applications are the top level, which can be mobile and web applications for solving a specific operational task of scientists, such as submission of reports or various research methods digitised in a convenient format.

It is necessary to ensure the deployment of private science clouds in Ukrainian data centres for national security purposes and to provide for a reserve of future computing capacity for national research.

6. Simplifying equipment procurement procedures

First, the procedure of tender procurement of unique scientific equipment for the activities of CSUSEs has to be simplified. It is advisable to consider adapting the best international practices of innovative procurement. To receive assistance from foreign partners, the mechanisms for obtaining used scientific equipment as humanitarian aid should be significantly simplified.

To expand the capabilities of HEIs/SIs, research groups of HEIs/SIs should be allowed to include ordering of services for the use of unique research-intensive equipment of CSUSEs in the estimates of research and development projects submitted for tenders arranged by MESU/NASU, as an alternative to the expenditures for purchasing materials/reagents/components. Provide universal explanations for all RIs on the calculation of depreciation charges in order to accumulate capital expenditures for equipment modernisation and renewal, as well as

ensure mandatory operating expense items for consumables to maintain equipment in working order in accordance with the technical documentation of the equipment.

7. Adjustment of regional smart specialisation strategies in the context of post-war recovery to create competitive value chains based on local RI

The following steps will help to practically integrate RI issues into regional smart specialisation strategies:

1. analysis of the current state of RI in the region and identification of industries where RI can provide competitive advantages, considering the needs of the region;
2. engaging HEIs/SIs that have RI in the discussion and development of smart specialisation strategies;
3. when determining the priorities of smart specialisation for the region, it is necessary to consider RI as a key factor in achieving these goals;
4. the regional smart specialisation strategy should include initiatives and activities aimed at developing RI in the selected smart priorities;
5. when looking for resources to implement initiatives and activities aimed at developing RI in the selected smart priorities, it is necessary to consider opportunities for cooperation with other regions or countries;
6. conducting awareness-raising campaigns to advertise the activities and capabilities of RI.

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